PRIMITIVE RECURSIVE ARITHMETIC AND METAMATHEMATICS

1. Introduction

In these notes we develop primitive recursive arithmetic. Although our development can be fully formalized in the first-order logic, we adopt a style analogous to set theory textbooks: rigorous but expressed using natural-language arguments rather than syntactical intricacies. The main difference is that, unlike set theory, our universe of discourse consists of natural numbers rather than sets. Consequently, all the theories we introduce share the feature that the range of their quantifiers is the set of natural numbers.

2. First-order logic

We denote object-language variables by

$$n, m, k, ...$$

In these notes variables range over *natural numbers*. We also use

$$\mathrm{n}, \mathrm{m}, \mathrm{k}, ...$$

to denote natural numbers at meta-level. For example we write $\vec{n}$ for a tuple $(n_1, ..., n_{\mathrm{n}})$ of object-language variables of length n, where n is some meta-level natural number.

We operate within classical deductive system introduced in [Monygham, 2026]. Free variables in formulas are treated as parameters. No implicit universal closure is ever assumed. We implicitly use alphabetic variants and do not bother with the usual issues related to substitution. Moreover, we apply conservativity results described in [Monygham, 2026]. Specifically, if $\mathcal{L}$ is a first-order language and Γ is a collection of formulas of $\mathcal{L}$, then we freely extend $\mathcal{L}$ and Γ as follows.

1. For every n-tuple $\vec{n}$ of variables and a formula $\Phi(\vec{n})$ we may add a fresh n-ary relation symbol R to $\mathcal{L}$ and extend Γ by adding

$$\forall_{\vec{n}} \ (\Phi(\vec{n}) \Leftrightarrow R(\vec{n}))$$

2. For every n-tuple $\vec{n}$ of variables and a formula $\Phi(\vec{n}, m)$ such that

$$\Gamma \vdash \exists_m \Phi(\vec{n}, m)$$

we may add a fresh constant symbol c to $\mathcal{L}$ and extend Γ by adding $\Phi(\vec{n}, c)$.

In the next section we introduce system PRA of primitive recursive arithmetic, but we want to inform the reader about important convention that we adopt.

1. Results labeled by PRA denote a statement derivable in PRA and the accompanying proof is intended to be in PRA.
2. Results labeled by PRA-**schema** denote a schema of statements derivable in PRA and the accompanying proof is intended to be an effective procedure that takes meta-theoretic data as inputs and produces a derivation in PRA.
3. Results labeled by **Metatheory** are meta-theoretic.

3. THE SYSTEM OF PRIMITIVE RECURSIVE ARITHMETIC

In this section we introduce the system PRA. The language of PRA consists only of function symbols. It contains base symbols

Zero symbols.

$\underbrace{\mathbf{Z}_k}_{k-ary}$ for every arity k. Also $\mathbf{Z}_0$ is denoted by 0.

The successor.

$\underbrace{\mathbf{S}}_{unary}$

Projection symbols.

$\underbrace{\mathbf{I}^i_k}_{k-ary}$ for every nonzero arity k and $1 \leq i \leq k$.

and is defined inductively by the following operations.

Composition.

If G is a k-ary symbol and $F_1, ..., F_k$ are m-ary symbols, then $\mathrm{comp}(G, F_1, ..., F_k)$ is a m-ary symbol. We say that $\mathrm{comp}(G, F_1, ..., F_k)$ is obtained from $G, F_1, ..., F_k$ by *composition*.

Primitive recursion.

If G is $(k+2)$-ary symbol and H is a k-ary symbol, then $\mathrm{prim}(G, H)$ is a $(k+1)$-ary symbol. We say that $\mathrm{prim}(G, H)$ is obtained from G, H by *primitive recursion*.

Definition 3.1. Function symbols of the language of PRA are called *primitive recursive* functions.

The nonlogical axioms of PRA are universal closures of the following formulas.

1. $\mathbf{Z}_k(\vec{n}) = 0$ for k-tuple $\vec{n}$ of variables.
2. $\mathbf{S}(n) \neq 0$
3. $\mathbf{I}^i_k(\vec{n}) = n_i$ for k-tuple $\vec{n}$ of variables and all $1 \leq i \leq k$.
4. If G is a k-ary symbol and $F_1, ..., F_k$ are m-ary symbols, then
$$\mathrm{comp}(G, F_1, ..., F_k)(\vec{n}) = G\left(F_1(\vec{n}), ..., F_k(\vec{n})\right)$$
for m-tuple $\vec{n}$ of variables.
5. If G is $(k+2)$-ary symbol and H is a k-ary symbol, then
$$\mathrm{prim}(G, H)(\vec{n}, 0) = H(\vec{n})$$
and
$$\mathrm{prim}(G, H)\left(\vec{n}, S(m)\right) = G\left(\vec{n}, m, \mathrm{prim}(G, H)(\vec{n}, m)\right)$$
where $\vec{n}$ is a k-tuple of variables and n is a variable distinct from $\vec{n}$.
6. **Induction axiom schema.**
Suppose that $\Phi(\vec{n}, m)$ is a quantifier-free formula in the language of PRA. Then
$$\left(\Phi(\vec{n}, 0) \wedge \forall_m \left(\Phi(\vec{n}, m) \Rightarrow \Phi(\vec{n}, S(m))\right)\right) \Rightarrow \forall_m \Phi(\vec{n}, m)$$

In addition to this nonlogical axioms PRA has usual first-order axioms with identity.

From now on unless otherwise stated formulas are always understood to be formulas of the language of PRA.

4. Rudimentary results in primitive recursive arithmetic

The reader may find this section rather dry and tedious. The goal here is to show that the most rudimentary facts about natural numbers can be proved in PRA. First we introduce familiar notation.

Definition 4.1. $1 = \mathbf{S}(0), 2 = \mathbf{S}(\mathbf{S}(0))$

Definition 4.2. $\mathbf{P}(0) = 0$, $\mathbf{P}(\mathbf{S}(n)) = n$.

Corollary 4.3 [PRA]. *The function* $\mathbf{S}$ *is injective.*

Proof. If $\mathbf{S}(n) = \mathbf{S}(m)$, then $n = \mathbf{P}(\mathbf{S}(n)) = \mathbf{P}(\mathbf{S}(m)) = m$. □

Corollary 4.4 [PRA]. *If* $n \neq 0$*, then there exists* m *such that* $\mathbf{S}(m) = n$.

Proof. We prove the assertion by induction on n. For $n = 0$ it is clear. Suppose that it holds for some n. Then

$$\mathbf{S}\left(\mathbf{P}\left(\mathbf{S}(n)\right)\right) = \mathbf{S}(n)$$

and hence it holds for $\mathbf{S}(n)$. Thus formula in question holds for all n. □

Next we introduce addition and multiplication.

Definition 4.5. $n + 0 = n$, $n + \mathbf{S}(m) = \mathbf{S}(n + m)$

Definition 4.6. $n \cdot 0 = 0$, $n \cdot \mathbf{S}(m) = n \cdot m + n$

Theorem 4.7 [PRA]. *The following assertions hold.*

(1) $n + m = m + n$

(2) $(n + m) + k = n + (m + k)$

(3) $n + k = m + k \Rightarrow n = m$

(4) $n + m = 0 \Rightarrow (n = 0 \wedge m = 0)$

(5) $n \cdot (m + k) = n \cdot m + n \cdot k$

(6) $n \cdot m = m \cdot n$

(7) $(n \cdot m) \cdot k = n \cdot (m \cdot k)$

(8) $n \cdot m = 0 \Rightarrow (n = 0 \vee m = 0)$

Proof. We leave the proof as an exercise. □

Now we introduce ordering relation.

Definition 4.8. $n \leq m \equiv \exists_k\, m = k + n$

Definition 4.9. $n < m \equiv n \leq m \wedge n \neq m$

In order to prove that ordering relation is quantifier-free we need truncated subtraction function.

Definition 4.10. $n \dot{-} 0 = n$, $n \dot{-} \mathbf{S}(m) = \mathbf{P}\left(n \dot{-} m\right)$

Proposition 4.11 [PRA]. *The following assertions are equivalent.*

(i) $n \leq m$

(ii) $m = (m \dot{-} n) + n$

For the proof we need the following result.

Lemma 4.11.1 [PRA]. *The following assertions hold.*

(1) $\mathbf{S}(m) \dot{-} \mathbf{S}(n) = m \dot{-} n$

(2) $(k + n) \dot{-} n = k$ *for every* n.

Proof of the lemma. The proof of **(1)** goes by induction on n. For base case we have

$$\mathbf{S}(m) \dot{-} \mathbf{S}(0) = \mathbf{P}(\mathbf{S}(m) \dot{-} 0) = \mathbf{P}(\mathbf{S}(m)) = m = m \dot{-} 0$$

Suppose that the assertion holds for some n. Then

$$\mathbf{S}(m) \dot{-} \mathbf{S}(\mathbf{S}(n)) = \mathbf{P}(\mathbf{S}(m) \dot{-} \mathbf{S}(n)) \underbrace{=}_{\text{induction hypothesis}} \mathbf{P}(m \dot{-} n) = m \dot{-} \mathbf{S}(n)$$

This proves that **(1)** holds.

The proof of **(2)** goes by induction on n. The base case is $k \dot{-} 0 = k$ and this follows by definition of $\dot{-}$. Suppose that $(k + n) \dot{-} n = k$ for some n. Then

$$(k + \mathbf{S}(n)) \dot{-} \mathbf{S}(n) = \mathbf{S}(k + n) \dot{-} \mathbf{S}(n) \underbrace{=}_{\textbf{(1)}} (k + n) \dot{-} n \underbrace{=}_{\text{induction hypothesis}} k$$

This completes the proof. □

Proof of the proposition. Suppose that **(i)** holds. Then there exists k such that $m = k + n$. We have

$$k \underbrace{=}_{\text{Lemma 4.11.1}} (k + n) \dot{-} n = m \dot{-} n$$

and hence $m = (m \dot{-} n) + n$. This is **(ii)**.

The implication **(ii)** $\Rightarrow$ **(i)** is obvious. □

Theorem 4.12 [PRA]. *The following assertions hold.*

(1) $0 \leq n$

(2) $n \leq n$

(3) $n < m \Leftrightarrow \exists_k (k \neq 0 \wedge m = k + n)$

(4) $(n \leq m \wedge m \leq k) \Rightarrow n \leq k$

(5) $(n \leq m \vee m \leq n)$

(6) $n \leq m \Rightarrow n \cdot k \leq m \cdot k$

(7) $n \leq m \Rightarrow n + k \leq m + k$

Proof. According to Proposition 4.11 formula $n \leq m$ is equivalent to a quantifier-free formula. We only prove assertion **(5)** and leave the rest for the reader. Fix n as a parameter and prove by induction on m that $n \leq m$ or $m \leq n$. Note that $0 \leq n$ according to **(1)**. This proves the base case. Suppose that the assertion holds for some m. We consider cases.

- If $n \leq m$, then there exists k such that $m = k + n$. Note that

  $$\mathbf{S}(m) = \mathbf{S}(k + n) \underbrace{=}_{\text{Theorem 4.7}} \mathbf{S}(n + k) = n + \mathbf{S}(k) \underbrace{=}_{\text{Theorem 4.7}} \mathbf{S}(k) + n$$

  and hence $n \leq \mathbf{S}(m)$.

- If $m = n$, then
$$\mathbf{S}(m) = \mathbf{S}(n) = n + 1 \underbrace{=}_{\text{Theorem 4.7}} 1 + n$$
and hence $n \leq \mathbf{S}(m)$.
- If $m \leq n$ and $m \neq n$, then by definition $m < n$. Hence by **(3)** there exists $k \neq 0$ such that $n = k + m$. Corollary 4.4 implies that there exists l such that $k = \mathbf{S}(l)$. Thus
$$n = k + m = \mathbf{S}(l) + m \underbrace{=}_{\text{Theorem 4.7}} m + \mathbf{S}(l) = \mathbf{S}(m + l) \underbrace{=}_{\text{Theorem 4.7}} \mathbf{S}(l + m) = l + \mathbf{S}(m)$$
This proves that $\mathbf{S}(m) \leq n$.

Thus $n \leq \mathbf{S}(m)$ or $\mathbf{S}(m) \leq n$. Hence **(5)** follows by induction on m. □

5. Primitive recursive formulas and characteristic functions

This section is devoted to important schemas in PRA.

Although it is very basic, it is worth noting, in order to make presentation rigorous that the composition of **S** and the zero function gives rise for every k to the constant k-ary function that admits value 1. By convention we denote this function by 1 and assume that its use is clear from the context.

Definition 5.1. $\mathrm{sign}(0) = 0$, $\mathrm{sign}(\mathbf{S}(n)) = 1$

Definition 5.2. Let $\Phi(\vec{n})$ be a formula and let C be a primitive recursive function. Suppose that
$$\forall_{\vec{n}} (C(\vec{n}) = 0 \vee C(\vec{n}) = 1) \wedge \forall_{\vec{n}} (\Phi(\vec{n}) \Leftrightarrow C(\vec{n}) = 1)$$
Then C is a *characteristic function* of $\Phi(\vec{n})$.

Theorem 5.3 [PRA-**schema**]. *Let $\Phi(\vec{n})$ be a quantifier-free formula. Then $\Phi(\vec{n})$ admits a characteristic function.*

Proof. Fix a quantifier-free formula $\Phi(\vec{n})$. For each subformula $\Psi(\vec{n})$ of $\Phi(\vec{n})$ we construct a characteristic function C_Ψ. The construction is a recursive procedure.

(1) $\Psi(\vec{n})$ is of the form $F_1(\vec{n}) = F_2(\vec{n})$ where F_1, F_2 are function symbols. Then we define
$$C_\Psi(\vec{n}) = 1 \dot{-} \mathrm{sign}\big(F_1(\vec{n}) \dot{-} F_2(\vec{n}) + F_2(\vec{n}) \dot{-} F_1(\vec{n})\big)$$
Clearly the right hand side is defined in terms of composition from sign, $+, \dot{-}, F_1, F_2$.

We prove that C_Ψ is a characteristic function for $\Psi(\vec{n})$. Clearly $C_\Psi(\vec{n}) = 0, 1$ for all $\vec{n}$. Moreover, by definition of $\dot{-}$ and sign the formula $C_\Psi(\vec{n}) = 1$ is equivalent to
$$F_1(\vec{n}) \dot{-} F_2(\vec{n}) + F_2(\vec{n}) \dot{-} F_1(\vec{n}) = 0$$
By Proposition 4.11 and Theorem 4.12 this last formula is provably equivalent in PRA to $F_1(\vec{n}) = F_2(\vec{n})$.

(2) $\Psi(\vec{n})$ is $\Psi_1(\vec{n}) \vee \Psi_2(\vec{n})$ for formulas $\Psi_1(\vec{n})$ and $\Psi_2(\vec{n})$ with characteristic functions $C_{\Psi_1}(\vec{n})$ and $C_{\Psi_2}(\vec{n})$, respectively. Then
$$C_\Psi(\vec{n}) = \mathrm{sign}\left(C_{\Psi_1}(\vec{n}) + C_{\Psi_2}(\vec{n})\right)$$
is a characteristic function of $\Psi(\vec{n})$.

(3) $\Psi(\vec{n})$ is $\Psi_1(\vec{n}) \wedge \Psi_2(\vec{n})$ for formulas $\Psi_1(\vec{n})$ and $\Psi_2(\vec{n})$ with characteristic functions $C_{\Psi_1}(\vec{n})$ and $C_{\Psi_2}(\vec{n})$, respectively. Then
$$C_\Psi(\vec{n}) = C_{\Psi_1}(\vec{n}) \cdot C_{\Psi_2}(\vec{n})$$
is a characteristic function of $\Psi(\vec{n})$.

(4) $\Psi(\vec{n})$ is $\neg\Theta(\vec{n})$ for some formula $\Theta(\vec{n})$ with characteristic function $C_\Theta(\vec{n})$. Then

$$C_\Psi(\vec{n}) = 1 \dot{-} C_\Theta(\vec{n})$$

is a characteristic function of $\Psi(\vec{n})$.

(5) $\Psi(\vec{n})$ is $\Psi_1(\vec{n}) \Rightarrow \Psi_2(\vec{n})$ for formulas $\Psi_1(\vec{n})$ and $\Psi_2(\vec{n})$ with characteristic functions $C_{\Psi_1}(\vec{n})$ and $C_{\Psi_2}(\vec{n})$, respectively. Since $\Psi_1(\vec{n}) \Rightarrow \Psi_2(\vec{n})$ is logically equivalent to $(\neg\Psi_1(\vec{n})) \vee \Psi_2(\vec{n})$ this case reduces to previously considered cases.

□

Definition 5.4. A formula $\Phi(\vec{n})$ that admits a characteristic function is *primitive recursive*.

Corollary 5.5 [PRA-**schema**]. *Let $\Phi(\vec{n})$ be a formula. Then the following assertions are equivalent.*

(i) *$\Phi(\vec{n})$ is primitive recursive.*

(ii) *There exists a quantifier-free formula $\Theta(\vec{n})$ such that*

$$\Phi(\vec{n}) \Leftrightarrow \Theta(\vec{n})$$

Proof. The implication **(i)** $\Rightarrow$ **(ii)** is straightforward.

The implication **(ii)** $\Rightarrow$ **(i)** follows from Theorem 5.3. □

Corollary 5.6 [PRA-**schema**]. *The collection of primitive recursive formulas is closed under $\wedge, \vee, \Rightarrow, \Leftrightarrow, \neg$.*

Proof. By Corollary-Schema 5.5 primitive recursive formulas are provably equivalent to quantifier-free formulas. By definition quantifier-free fornulas are closed under $\wedge, \vee, \Rightarrow, \Leftrightarrow$. Hence primitive recursive formulas are closed under $\wedge, \vee, \Rightarrow, \Leftrightarrow, \neg$. □

Corollary 5.7 [PRA-**schema**]. *Let F_i be a* k*-ary primitive recursive function for* $i = 1, ..., n$. *Suppose that $\vec{n}$ is a* k*-tuple of variables. Let $\Phi_i(\vec{n})$ be primitive recursive formulas for* $i = 1, ..., n$ *such that the following assertions hold.*

(1) $\forall_{\vec{n}}\big(\Phi_1(\vec{n}) \vee ... \vee \Phi_n(\vec{n})\big)$

(2) $\forall_{\vec{n}}\big(\neg\left(\Phi_i(\vec{n}) \wedge \Phi_j(\vec{n})\right)\big)$ *for all* $i \neq j$ *such that* $i, j = 1, ..., n$.

Then there exists k*-ary primitive recursive function F such that*

$$\forall_{\vec{n}}\big(\Phi_i(\vec{n}) \Rightarrow F(\vec{n}) = F_i(\vec{n})\big)$$

for every $i = 1, ..., n$.

Proof. According to Corollary 5.5 there exists a characteristic function C_i for $\Phi_i(\vec{n})$ for $i = 1, ..., n$. We define

$$F(\vec{n}) = F_1(\vec{n}) \cdot C_1(\vec{n}) + ... + F_n(\vec{n}) \cdot C_n(\vec{n})$$

Then F clearly satisfies the assertion. □

6. Cummulative operators and bounded quantifiers

We introduce two basic operators.

Definition 6.1. Let F be a $(\mathrm{k}+1)$-ary primitive recursive function. We define a *cummulative sum*

$$(\vec{n}, m) \mapsto \sum_{i<m} F(\vec{n}, i)$$

by primitive recursion

$$\sum_{i<0} F(\vec{n}, i) = 0, \sum_{i<\mathbf{S}(m)} F(\vec{n}, i) = F(\vec{n}, m) + \sum_{i<m} F(\vec{n}, i)$$

Definition 6.2. Let F be a $(\mathrm{k}+1)$-ary primitive recursive function. We define a *cummulative product*

$$(\vec{n}, m) \mapsto \prod_{i<m} F(\vec{n}, i)$$

by primitive recursion

$$\prod_{i<0} F(\vec{n}, i) = 1, \prod_{i<\mathbf{S}(m)} F(\vec{n}, i) = F(\vec{n}, m) \cdot \prod_{i<m} F(\vec{n}, i)$$

Remark 6.3. Let F be a $(\mathrm{k}+1)$-ary primitive recursive function. Then we use the following notation

$$\sum_{k\leq i<m} F(\vec{n}, i) \equiv \sum_{i<m\dot{-}k} F(\vec{n}, i+k), \prod_{k\leq i<m} F(\vec{n}, i) = \prod_{i<m\dot{-}k} F(\vec{n}, i+k)$$

Next result establishes basic properties of cummulative operators.

Fact 6.4 [PRA-**schema**]. *Let F, G be $(\mathrm{k}+1)$-ary primitive recursive functions.*

(1) $(k < m) \Rightarrow \sum_{i<m} F(\vec{n}, i) = \sum_{i<k} F(\vec{n}, i) + \sum_{k\leq i<m} F(\vec{n}, i)$

(2) $\sum_{i<m} F(\vec{n}, i) + \sum_{i<m} G(\vec{n}, i) = \sum_{i<m} (F(\vec{n}, i) + G(\vec{n}, i))$

(3) $(k < m) \Rightarrow \prod_{i<m} F(\vec{n}, i) = \prod_{i<k} F(\vec{n}, i) \cdot \prod_{k\leq i<m} F(\vec{n}, i)$

(4) $\prod_{i<m} F(\vec{n}, i) \cdot \prod_{i<m} G(\vec{n}, i) = \prod_{i<m} (F(\vec{n}, i) \cdot G(\vec{n}, i))$

Proof. Left for the reader as an exercise. □

Remark 6.5. We use the following notation

$$\forall_{i<m} \Phi(\vec{n}, i) \equiv \forall_i (i < m \Rightarrow \Phi(\vec{n}, i)), \exists_{i<m} \Phi(\vec{n}, i) \equiv \exists_i (i < m \wedge \Phi(\vec{n}, i))$$

for every formula $\Phi(\vec{n}, i)$.

Theorem 6.6 [PRA-**schema**]. *Let $\Phi(\vec{n}, i)$ be a primitive recursive formula. Then formulas*

$$\forall_{i<m} \Phi(\vec{n}, i), \exists_{i<m} \Phi(\vec{n}, i)$$

are primitive recursive.

For the proof we need the following results.

Lemma 6.6.1 [PRA-**schema**]. *Let F be a $(\mathrm{k}+1)$-ary primitive recursive function. Then*

$$\forall_{i<m} F(\vec{n}, i) > 0 \Leftrightarrow \forall_k \left(k \leq m \Rightarrow \prod_{i<k} F(\vec{n}, i) > 0 \right)$$

Proof of the lemma. Suppose that $F(\vec{n}, i) > 0$ for every $i < m$. We prove

$$\forall_k \left(k \leq m \Rightarrow \prod_{i<k} F(\vec{n}, i) > 0 \right)$$

by induction on k. The base case is straightforward. Suppose that $k < m$ and $\prod_{i<k} F(\vec{n}, i) > 0$. Then

$$\prod_{i<\mathbf{S}(k)} F(\vec{n}, i) = F(\vec{n}, k) \cdot \prod_{i<k} F(\vec{n}, i) \underbrace{\geq}_{\text{induction hypothesis}} F(\vec{n}, k) \cdot 1 = F(\vec{n}, k) > 0$$

and hence the assertion follows by induction.

Suppose that

$$\forall_k \left(k \leq m \Rightarrow \prod_{i<k} F(\vec{n}, i) > 0 \right)$$

holds. Pick $k < m$. Then

$$F(\vec{n}, k) \cdot \prod_{i<k} F(\vec{n}, i) = \prod_{i<\mathbf{S}(k)} F(\vec{n}, i) > 0$$

Thus $F(\vec{n}, k) > 0$. Therefore, $F(\vec{n}, i) > 0$ for every $i < m$. □

Lemma 6.6.2 [PRA-**schema**]. *Let F be a $(\mathrm{k}+1)$-ary primitive recursive function. Then*

$$\forall_{i<m} F(\vec{n}, i) = 0 \Leftrightarrow \forall_k \left(k \leq m \Rightarrow \sum_{i<k} F(\vec{n}, i) = 0 \right)$$

Proof of the lemma. Suppose that $F(\vec{n}, i) = 0$ for every $i < m$. We prove

$$\forall_k \left(k \leq m \Rightarrow \sum_{i<k} F(\vec{n}, i) = 0 \right)$$

by induction on k. The base case is straightforward. Suppose that $k < m$ and $\sum_{i<k} F(\vec{n}, i) = 0$. Then

$$\sum_{i<\mathbf{S}(k)} F(\vec{n}, i) = F(\vec{n}, k) + \sum_{i<k} F(\vec{n}, i) \underbrace{=}_{\text{induction hypothesis}} F(\vec{n}, k) = 0$$

and hence the assertion follows by induction.

Suppose that

$$\forall_k \left(k \leq m \Rightarrow \sum_{i<k} F(\vec{n}, i) = 0 \right)$$

holds. Pick $k < m$. Then

$$F(\vec{n}, k) + \sum_{i<k} F(\vec{n}, i) = \sum_{i<\mathbf{S}(k)} F(\vec{n}, i) = 0$$

Thus $F(\vec{n}, k) = 0$. Therefore, $F(\vec{n}, i) = 0$ for every $i < m$. □

Proof of the theorem. Let C be a characteristic function of $\Phi(\vec{n}, i)$.

According to Lemma 6.6.1 we have $\operatorname{sign}\left(\prod_{i<m} C(\vec{n}, i) \right) = 1$ if and only if $C(\vec{n}, i) = 1$ for all $i < m$. Thus

$$(\vec{n}, m) \mapsto \operatorname{sign}\left(\prod_{i<m} C(\vec{n}, i) \right)$$

is a characteristic function of $\forall_{i<m} \Phi(\vec{n}, i)$.

According to Lemma 6.6.2 we have $\operatorname{sign}\left(\sum_{i<m} C(\vec{n},i)\right) = 1$ if and only if there exists $i < m$ such that $C(\vec{n},i) = 1$. Thus

$$(\vec{n},m) \mapsto \operatorname{sign}\left(\sum_{i<m} C(\vec{n},i)\right)$$

is a characteristic function of $\exists_{i<m}\Phi(\vec{n},i)$. □

7. Induction schemata and bounded minimum

Next theorem-schema establishes equivalence of schemata expressing versions of mathematical induction for primitive recursive formulas.

Theorem 7.1 [PRA-**schema**]. *The following schemata for primitive recursive formulas hold in* PRA.

(i) Induction schema.

$$\left(\Phi(\vec{n},0) \wedge \forall_m\left(\Phi(\vec{n},m) \Rightarrow \Phi(\vec{n},\mathbf{S}(m))\right)\right) \Rightarrow \forall_m \Phi(\vec{n},m)$$

(ii) Complete induction schema.

$$\forall_m\left(\forall_{i<m}\Phi(\vec{n},i) \Rightarrow \Phi(\vec{n},m)\right) \Rightarrow \forall_m \Phi(\vec{n},m)$$

(iii) Minimum principle.

$$\exists_m \Phi(\vec{n},m) \Rightarrow \exists_k\left(\Phi(\vec{n},k) \wedge \forall_m\left(\Phi(\vec{n},m) \Rightarrow k \leq m\right)\right)$$

(iv) Finite descent principle.

$$\left(\exists_m \Phi(\vec{n},m) \wedge \forall_m\left(\left(\Phi(\vec{n},m) \wedge m \neq 0\right) \Rightarrow \exists_{i<m}\Phi(\vec{n},i)\right)\right) \Rightarrow \Phi(\vec{n},0)$$

Proof. Corollary 5.5 implies that **(i)** holds in PRA. It suffices to prove that all schemata in the statement are equivalent in PRA.

Suppose that **(i)** holds. Pick some primitive recursive formula $\Phi(\vec{n},i)$. Define a formula

$$\Psi(\vec{n},m) \equiv \forall_{i<m}\Phi(\vec{n},i)$$

Then $\Psi(\vec{n},m)$ is primitive recursive according to Theorem 6.6. Next assuming

$$\forall_m\left(\forall_{i<m}\Phi(\vec{n},i) \Rightarrow \Phi(\vec{n},m)\right)$$

we derive

$$\Psi(\vec{n},0) \wedge \forall_m\left(\Psi(\vec{n},m) \to \Psi(\vec{n},\mathbf{S}(m))\right)$$

in PRA. Since **(i)** holds, we infer that $\forall_m \Psi(\vec{n},m)$. Thus $\forall_m \forall_{i<m}\Phi(\vec{n},i)$. Combining it with

$$\forall_m\left(\forall_{i<m}\Phi(\vec{n},i) \Rightarrow \Phi(\vec{n},m)\right)$$

and logical axiom

$$\forall_m\left(\forall_{i<m}\Phi(\vec{n},i) \Rightarrow \Phi(\vec{n},m)\right) \Rightarrow \left(\forall_m \forall_{i<m}\Phi(\vec{n},i) \Rightarrow \forall_m \Phi(\vec{n},m)\right)$$

we infer that $\forall_m \Phi(\vec{n},m)$. This proves that schema **(i)** implies schema **(ii)** in PRA.

Fix a primitive recursive formula $\Phi(\vec{n},m)$ and suppose that $\exists_m \Phi(\vec{n},m)$. Then it is not the case $\forall_m(\neg\Phi(\vec{n},m))$. Note that $\neg\Phi(\vec{n},m)$ is primitive recursive by Corollary 5.6. If **(ii)** holds for $\neg\Phi(\vec{n},m)$, then

$$\forall_m\left(\forall_{i<m}\left(\neg\Phi(\vec{n},i)\right) \Rightarrow \left(\neg\Phi(\vec{n},m)\right)\right)$$

is not the case. Let k be such that

$$\left(\neg\left(\neg\Phi(\vec{n},k)\right)\right) \wedge \forall_{i<k}\left(\neg\Phi(\vec{n},i)\right)$$

Then

$$\Phi(\vec{n},k) \wedge \forall_{i<k}\left(\neg\Phi(\vec{n},i)\right)$$

Since

$$\forall_{i<k}\left(\neg\Phi(\vec{n},i)\right) \equiv \forall_i\left(k \leq i \vee \left(\neg\Phi(\vec{n},i)\right)\right)$$

and

$$\left(k \leq i \vee \left(\neg\Phi(\vec{n},i)\right)\right) \Leftrightarrow \left(\Phi(\vec{n},i) \Rightarrow k \leq i\right)$$

is a theorem of classical logic, we derive that

$$\Phi(\vec{n},k) \wedge \forall_m\left(\Phi(\vec{n},m) \Rightarrow k \leq m\right)$$

This completes the proof that schema **(ii)** implies schema **(iii)**.

Next suppose that **(iii)** holds. Let $\Phi(\vec{n},m)$ be a primitive recursive formula such that

$$\exists_m \Phi(\vec{n},m) \wedge \forall_m\left(\left(\Phi(\vec{n},m) \wedge m \neq 0\right) \Rightarrow \exists_{k<m}\Phi(\vec{n},k)\right)$$

Note that $\exists_m \Phi(\vec{n},m)$ and **(iii)** imply that there exists k such that

$$\Phi(\vec{n},k) \wedge \forall_m\left(\Phi(\vec{n},m) \Rightarrow k \leq m\right)$$

Assume that $k \neq 0$. Then $\forall_m\left(\left(\Phi(\vec{n},m) \wedge m \neq 0\right) \Rightarrow \exists_{i<m}\Phi(\vec{n},i)\right.$ implies that there exists $i < k$ such that $\Phi(\vec{n},i)$. Since $\forall_m\left(\Phi(\vec{n},m) \Rightarrow k \leq m\right)$, we derive that $k \leq i$. Together with PRA this implies that $k < k$ and this is a contradiction. Thus $k = 0$ and

$$\left(\exists_m \Phi(\vec{n},m) \wedge \forall_m\left(\left(\Phi(\vec{n},m) \wedge m \neq 0\right) \Rightarrow \exists_{i<m}\Phi(\vec{n},i)\right)\right) \Rightarrow \Phi(\vec{n},0)$$

Hence schema **(iii)** implies schema **(iv)** in PRA.

Finally assume that schema **(iv)** holds. Suppose that $\Phi(\vec{n},m)$ is a primitive recursive formula such that

$$\Phi(\vec{n},0) \wedge \forall_m\left(\Phi(\vec{n},m) \Rightarrow \Phi(\vec{n},\mathbf{S}(m))\right)$$

Suppose that $\neg\Phi(\vec{n},m)$ for some $m \neq 0$. Then PRA proves that there exists l such that $m = \mathbf{S}(l)$. It follows that $\Phi(\vec{n},l) \Rightarrow \Phi(\vec{n},\mathbf{S}(l))$. Combining this with $\neg\Phi(\vec{n},m)$ we infer that $\neg\Phi(\vec{n},l)$. Again invoking PRA we obtain

$$\left(\left(\neg\Phi(\vec{n},m)\right) \wedge m \neq 0\right) \Rightarrow \exists_{i<m}\left(\neg\Phi(\vec{n},i)\right)$$

By variable generalization

$$\forall_m\left(\left(\left(\neg\Phi(\vec{n},m)\right) \wedge m \neq 0\right) \Rightarrow \exists_{i<m}\left(\neg\Phi(\vec{n},i)\right)\right)$$

Suppose now that $\exists_m\left(\neg\Phi(\vec{n},m)\right)$. Then

$$\exists_m\left(\neg\Phi(\vec{n},m)\right) \wedge \forall_m\left(\left(\left(\neg\Phi(\vec{n},m)\right) \wedge m \neq 0\right) \Rightarrow \exists_{i<m}\left(\neg\Phi(\vec{n},i)\right)\right)$$

Since the negation of $\Phi(\vec{n},m)$ is primitive recursive by Corollary 5.6 and **(iv)** holds, we derive that $\neg\Phi(\vec{n},0)$. This leads to contradiction with $\Phi(\vec{n},0)$. Thus it is not the case that $\exists_m\left(\neg\Phi(\vec{n},m)\right)$ and hence $\forall_m \Phi(\vec{n},m)$. We deduce that schema **(iv)** implies schema **(i)** in PRA. □

Theorem 7.2 [PRA-**schema**]. *Let $\Phi(\vec{n},i)$ be a primitive recursive formula. Then there exists a primitive recursive function*

$$(\vec{n},m) \mapsto \mu_{i<m}\left[\Phi(\vec{n},i)\right]$$

such that the following assertions hold.

(1) $\exists_{i<m}\Phi(\vec{n},i) \Rightarrow \left(\Phi\left(\vec{n},\mu_{i<m}\left[\Phi(\vec{n},i)\right]\right) \wedge \forall_k\left(\Phi(\vec{n},k) \Rightarrow \mu_{i<m}\left[\Phi(\vec{n},i)\right] \leq k\right)\right)$

(2) $\left(\neg\exists_{i<m}\Phi(\vec{n},i)\right) \Rightarrow \mu_{i<m}\left[\Phi(\vec{n},i)\right] = m$

Proof. Note that formulas

$$\Psi(\vec{n},k) = \forall_{i<k}\,(\neg\Phi(\vec{n},i)) \wedge \Phi(\vec{n},k),\ \Theta(\vec{n},m) = \forall_{i<m}\,(\neg\Phi(\vec{n},i))$$

are primitive recursive by Corollary 5.6 and Theorem 6.6. Let C_Ψ, C_Θ be characteristic functions of $\Psi(\vec{n},k)$ and $\Theta(\vec{n},m)$, respectively. Then we set

$$\mu_{i<m}\big[\Phi(\vec{n},i)\big](\vec{n},m) = m \cdot C_\Theta(\vec{n},m) + \sum_{k<m} k \cdot C_\Psi(\vec{n},k)$$

Clearly $\mu_{i<m}\big[\Phi(\vec{n},i)\big]$ is primitive recursive.

We work in PRA and prove **(1)** and **(2)**.

Suppose that there exists $i < m$ such that $\Phi(\vec{n},i)$. According to the minimum principle (Theorem 7.1) there exists the smallest j such that $\Phi(\vec{n},j)$. Clearly $j < i$. Note that $\Psi(\vec{n},k)$ does not hold for all $k < j$ and $j < k$. Moreover, $\Theta(\vec{n},m)$ does not hold. Hence

$$\mu_{i<m}\big[\Phi(\vec{n},i)\big](\vec{n},m) = m \cdot C_\Theta(\vec{n},m) + \sum_{k<m} k \cdot C_\Psi(\vec{n},k) = j \cdot C_\Psi(\vec{n},j) = j$$

This completes the proof of **(1)**.

Next assume that $\Phi(\vec{n},i)$ does not hold for all $i < m$. Then $\Psi(\vec{n},k)$ does not hold for all $k < m$ and $\Theta(\vec{n},m)$ holds. Hence

$$\mu_{i<m}\big[\Phi(\vec{n},i)\big](\vec{n},m) = m \cdot C_\Theta(\vec{n},m) + \sum_{k<m} k \cdot C_\Psi(\vec{n},k) = m \cdot C_\Theta(\vec{n},m) = m$$

This proves **(2)**. □

Definition 7.3. Let $\Phi(\vec{n},i)$ be a primitive recursive formula. Then $\mu_{i<m}\big[\Phi(\vec{n},i)\big]$ is the *bounded minimum* of $\Phi(\vec{n},i)$.

8. Elementary number theory in Primitive Recursive Arithmetic

In this section we develop substantial amount of elementary number theory in PRA.

Definition 8.1.

$$n|m \equiv \exists_k\, m = k \cdot n$$

If $n|m$, then n *divides* m or n is a *divisor* of m.

Fact 8.2 [PRA]. $n|m \Leftrightarrow \exists_{k\leq m}\, m = k \cdot n$

Proof. Left for the reader as an exercise. □

Definition 8.3.

$$\mathrm{Prime}(n) \equiv 1 < n \wedge \forall_m \big(m|n \Rightarrow (m = 1 \vee m = n)\big)$$

If $\mathrm{Prime}(n)$, then n is a *prime*.

Fact 8.4 [PRA]. $\mathrm{Prime}(n) \Leftrightarrow 1 < n \wedge \forall_{m<n}\ (m|n \Rightarrow m = 1)$

Proof. Left for the reader as an exercise. □

Proposition 8.5 [PRA]. *Let n be greater than* 1. *Then there exists prime m such that $m|n$.*

Proof. Fix $1 < n$. By Fact 8.2 formula

$$1 < m \wedge m|n$$

is primitive recursive. Since $1 < n \wedge n|n$, we apply the minimum principle to deduce that there exists the smallest m such that $1 < m \wedge m|n$. It suffices to prove that m is prime. Suppose that k is a divisor of m. If $1 < k < m$, then $k|n$ and this contradicts the minimality of m. Thus either $k = 1$ or $k = m$. □

Before proving the next very important theorem we need to introduce new primitive recursive function.

Definition 8.6. $0! = 1$, $(n+1)! = (n+1) \cdot n!$

Theorem 8.7 [PRA][Euclid]. *There exists primitive recursive function $n \mapsto p_n$ such that the following assertions hold.*

- **(1)** *p_n is prime for every n.*
- **(2)** *$p_n < p_{n+1}$ for every n.*
- **(3)** *$n < p_n$ for every n.*
- **(4)** *For every prime m there exists n such that $p_n = m$.*

We first need to prove the following results.

Lemma 8.7.1 [PRA]. $1 \leq m \leq n \Rightarrow m|n!$

Proof of the lemma. We prove this by induction on n. The base case is clear. Suppose that the result holds for some n. Pick $m \leq n+1$. If $m \leq n$, then $m|n!$ by induction hypothesis and hence

$$m|(n+1) \cdot n! = (n+1)!$$

If $m = n+1$, then $(n+1)! = m \cdot n!$. □

Lemma 8.7.2 [PRA]. *There exists a primitive recursive function G such that $G(n)$ is the smallest prime number greater than n.*

Proof of the lemma. According to Proposition 8.5 there exists prime m such that $m|n! + 1$. Clearly $m \leq n! + 1$. If $m \leq n$, then $1 \leq m \leq n$ and by Lemma 8.7.1 we derive that $m|n!$. Hence

$$m|(n! + 1) \dot{-} n! = 1$$

This is contradiction. Thus $n < m \leq n! + 1$. Next we define

$$G(n) = \mu_{i<n!+2}\big[\mathrm{Prime}(i) \wedge n < i\big]$$

According to Theorem 7.2 it follows that G is primitive recursive and $G(n)$ is the smallest prime number greater than n. □

Proof of the theorem. We define

$$p_0 = 2,\ p_{n+1} = G(p_n)$$

where G is primitive recursive function from Lemma 8.7.2. Since $n \mapsto p_n$ is obtained by primitive recursion, we derive that it is primitive recursive function. Clearly **(1)**, **(2)** and **(3)** hold.

Now we prove **(4)**. Let m be a prime number. Since $m < p_m$, there exists the smallest $k \leq m$ such that $m < p_k$. Note that $k \neq 0$ and hence there exists n such that $k = n+1$. Then $p_n \leq m < p_{n+1}$. If $p_n < m$, then

$$p_{n+1} = G(p_n) \leq m < p_{n+1}$$

and hence contradiction. Thus $m = p_n$. This proves **(4)**. □

Next we prove Euclidean division theorem in PRA.

Definition 8.8. $\operatorname{div}(n,m) = \mu_{i<n}\big[n < (i+1)\cdot m\big]$

Definition 8.9. $\operatorname{rem}(n,m) = n \dot{-} \operatorname{div}(n,m)\cdot m$

Theorem 8.10 [PRA][Euclidean division theorem]. *If $m \neq 0$, then*

$$n = \operatorname{div}(n,m)\cdot m + \operatorname{rem}(n,m)$$

and $\operatorname{rem}(n,m) < m$. Moreover, $\operatorname{div}(n,m)$ and $\operatorname{rem}(n,m)$ are unique with respect to these properties.

Proof. For $m \neq 0$ we have $n < (n+1)\cdot m$ and hence there exists $i < n+1$ such that $n < (i+1)\cdot m$. According to Theorem 7.2 the smallest i such that $n < (i+1)\cdot m$ is $\operatorname{div}(n,m)$. It follows that

$$\operatorname{div}(n,m)\cdot m \leq n < (\operatorname{div}(n,m)+1)\cdot m$$

Hence $n = \operatorname{div}(n,m)\cdot m + \operatorname{rem}(n,m)$ and $\operatorname{rem}(n,m) < m$.

Assume that

$$q_1\cdot m + r_1 = q_2\cdot m + r_2$$

for some $r_1, r_2 < m$. Without loss of generality we may assume that $r_1 \leq r_2$. Then there exists r such that $r_2 = r + r_1$. Hence $q_1\cdot m = q_2\cdot m + r$. It follows that $q_2 \leq q_1$ and there exists q such that $q_1 = q_2 + q$. Then $q\cdot m = r$. If $q \neq 0$, then

$$m \leq q\cdot m = r \leq r_2 < m$$

This is a contradiction. Thus $q = 0$ and consequently $q_1 = q_2$ and $r_1 = r_2$. □

Now we prove Bézout theorem and Chinese remainder theorem in PRA. Both these results are related to the notion of relatively prime numbers.

Definition 8.11.

$$n \perp m \equiv (n \neq 0 \wedge m \neq 0) \wedge \forall_k\big((k|n \wedge k|m) \Rightarrow k = 1\big)$$

If $n \perp m$, then n, m are *relatively prime*.

Fact 8.12 [PRA]. $n \perp m \Leftrightarrow \Big((n \neq 0 \wedge m \neq 0) \wedge \forall_{k\leq n}\big((k|n \wedge k|m) \Rightarrow k = 1\big)\Big)$

Proof. Left for the reader as an exercise. □

Theorem 8.13 [PRA][Bézout theorem]. *Let n, m be relatively prime. Then there exist a, b such that*

$$a\cdot n = b\cdot m + 1$$

The proof of Bézout theorem relies on two lemmas.

Lemma 8.13.1 [PRA]. *Let n, m be nonzero and let z be fixed. The following assertions are equivalent.*

- **(i)** *There exist a, b such that $a\cdot n = b\cdot m + k$.*
- **(ii)** *There exist a, b such that $a\cdot n + k = b\cdot m$.*

Proof of the lemma. Left for the reader as an exercise. □

Lemma 8.13.2 [PRA]. *Let n, m be nonzero and let k be smaller then n, m. Then the following assertions are equivalent.*

- **(i)** *There exist a, b such that $a\cdot n = b\cdot m + k$.*

(ii) *There exist $a < m$ and $b < n$ such that $a \cdot n = b \cdot m + k$.*

Proof of the lemma. Fix nonzero n, m and k not greater than n, m. Suppose that $a \cdot n = b \cdot m + k$. By Euclidean division theorem

$$a = q_a \cdot m + r_a,\ b = q_b \cdot n + r_b$$

for some q_a, q_b and $r_a < m, r_b < n$. Thus

$$q_a \cdot (n \cdot m) + r_a \cdot n = q_b \cdot (n \cdot m) + r_b \cdot m + k$$

Note that both $r_a \cdot n$ and $r_b \cdot m + k$ are smaller than $n \cdot m$. According to the uniqueness assertion of Euclidean division theorem we derive that $q_a = q_b$ and $r_a \cdot n = r_b \cdot m + k$. This completes the proof according to $r_a < m, r_b < n$. □

Proof of the theorem. We fix relatively prime n, m. Note that by definition n, m are nonzero.

Euclidean division theorem implies that there exists $k > 0$ smaller than n, m such that either

$$a \cdot n = b \cdot m + k$$

or

$$a \cdot n + k = b \cdot m$$

for some a, b. By Lemma 8.13.1 it follows that there exists $k > 0$ smaller than n, m such that

$$a \cdot n = b \cdot m + k$$

for some a, b. Applying the minimum principle and Lemma 8.13.2 we infer that there exists the smallest $k > 0$ such that $a \cdot n = b \cdot m + k$ for some a, b.

By Euclidean division algorithm there are q, r such that $m = q \cdot k + r$ and $r < k$. Then

$$(q \cdot a) \cdot n + r = (q \cdot b) \cdot m + q \cdot k + r = (q \cdot b + 1) \cdot m$$

Lemma 8.13.1 implies that $a \cdot n = b \cdot m + r$ for some a, b. If $0 < r$, then we obtain a contradiction with the minimality of k. Hence r is zero and k divides m.

Similarly by Euclidean division algorithm there are q, r such that $n = q \cdot k + r$ and $r < k$. Moreover, by Lemma 8.13.1 it follows that $a \cdot n + k = b \cdot m$ for some a, b. Then

$$(q \cdot b) \cdot m + r = (q \cdot a) \cdot n + q \cdot k + r = (q \cdot a + 1) \cdot n$$

This shows that $a \cdot n = b \cdot m + r$ for some a, b. If $r > 0$, then we obtain a contradiction with the minimality of k. Hence r is zero and k divides n.

Thus k divides both n and m. Since n and m are relatively prime, we infer that $k = 1$ and hence $a \cdot n = b \cdot m + 1$ for some a, b. □

Corollary 8.14 [PRA][Euclid's lemma]. $(n \perp m \wedge n \perp k) \Rightarrow n \perp m \cdot k$

Proof. According to Bézout theorem 8.13 there exist a, b, c, d such that $a \cdot m = b \cdot n + 1$ and $c \cdot k = d \cdot n + 1$. Then

$$(a \cdot c) \cdot (m \cdot k) = (b \cdot d + b + d) \cdot n + 1$$

Hence any common divisor of $m \cdot k$ and n divides 1. Thus n and $m \cdot k$ are relatively prime. □

Theorem 8.15 [PRA-**schema**][Chinese remainder theorem]. *Let F, G be primitive recursive functions. Suppose that the following assertions hold.*

(1) *$F(i)$ are pairwise relatively prime for all $i < m$*

(2) *$G(i) < F(i)$ for all $i < m$.*

Then there exists $n < \prod_{i<m} F(i)$ *such that*

$$\text{rem}\,(n, F(i)) = G(i)$$

for all $i < m$.

For the proof we need the following result

Lemma 8.15.1 [PRA-**schema**]. *Let* F *be a primitive recursive function and let* n *be such that* $n \perp F(i)$ *for all* $i < m$. *Then*

$$n \perp \prod_{i<m} F(i)$$

Proof of the lemma. The proof goes by induction on m by means of Corollary 8.14. The details are left for the reader. □

Proof of the theorem. Note that the statement can be expressed by primitive recursive formula. We apply induction on m. The base case is straightforward. Next suppose that

(1) $F(i)$ are pairwise relatively prime for all $i < m+1$

(2) $G(i) < F(i)$ for all $i < m+1$.

Then by induction hypothesis there exists k such that $k < \prod_{i<m} F(i)$ and

$$\text{rem}\,(k, F(i)) = G(i)$$

for all $i < m$. We set

$$\tilde{k} = \prod_{i<m+1} F(i) + k$$

Then

$$\text{rem}\,(\tilde{k}, F(i)) = G(i)$$

for all $i < m$ and $\tilde{k} \geq F(m) > G(m)$. By Lemma 8.15.1 and Bézout theorem 8.13 there exists a, b such that

$$a \cdot F(m) = b \cdot \prod_{i<m} F(i) + 1$$

We set

$$\tilde{n} = a \cdot F(m) \cdot (\tilde{k} \dot{-} G(m)) + G(m)$$

Then

$$\text{rem}\,(\tilde{n}, F(i)) = \text{rem}\left(b \cdot (\tilde{k} \dot{-} G(m)) \cdot \prod_{i<m} F(i) + G(m) + (\tilde{k} \dot{-} G(m)), F(i)\right) =$$

$$= \text{rem}\,\big(G(m) + (\tilde{k} \dot{-} G(m)), F(i)\big) \underbrace{=}_{\text{Proposition 4.11}} \text{rem}(\tilde{k}, F(i)) = G(i)$$

for all $i < m$. Moreover, we have

$$\text{rem}\,(\tilde{n}, F(m)) = \text{rem}\,\big(a \cdot F(m) \cdot (\tilde{k} \dot{-} G(m)) + G(m), F(m)\big) = \text{rem}(G(m), F(m)) = G(m)$$

Note that $n = \text{rem}\,\big(\tilde{x}, \prod_{i<m+1} F(i)\big)$ also satisfies these assertions and

$$n < \prod_{i<m+1} F(i)$$

This completes the proof. □

Finally, we prove fundamental theorem of arithmetic in PRA.

Definition 8.16. $n^0 = 1$, $n^{m+1} = n \cdot n^m$

Fact 8.17 [PRA]. *The following assertions hold.*

(1) $(n > 1 \wedge m < k) \Rightarrow n^m < n^k$

(2) $n > 1 \Rightarrow \forall_m\, m < n^m$

(3) $n^m \cdot n^k = n^{m+k}$

Proof. Left for the reader as an exercise. □

Definition 8.18. $\exp(n, m) = \mu_{i<n}\left[p_m^{i+1} \nmid n\right]$

First we note the following basic result.

Fact 8.19 [PRA]. $n \neq 0 \Rightarrow \left(p_m^i | n \Leftrightarrow i \leq \exp(n, m)\right)$

Proof. We have $n < p_m^n$ by Fact 8.17 and hence $p_m^n \nmid n$. Thus $\exp(n, m)$ is the smallest i such that $p_m^{i+1} \nmid n$. It follows that

$$p_m^{\exp(n,m)} | n,\ p_m^{\exp(n,m)+1} \nmid n$$

Next note that

(1) If $p_m^i | n$, then $p_m^j | n$ for every $j \leq i$.

(2) If $p_m^i \nmid n$, then $p_m^j \nmid n$ for every $i \leq j$.

The assertion now follows from the combination of these observations. □

Fact 8.20 [PRA]. $n \neq 0 \Rightarrow \forall_i \exp(n, i) < n$

Proof. According to Fact 8.19 we infer that $p_i^{\exp(n,i)} | n$. Since $n \neq 0$, we derive $p_i^{\exp}(n, i) \leq n$. Combine this with Fact 8.17 to deduce that

$$\exp(n, i) < p_i^{\exp(n,i)} \leq n$$

This completes the proof. □

Theorem 8.21 [PRA-**schema**][Unique prime factorization]. *The following assertions hold.*

(1) *If F is a primitive recursive function, then*

$$\exp\left(\prod_{i<m} p_i^{F(i)}, j\right) = \begin{cases} F(j) & \text{if } j < m \\ 0 & \text{if } m \leq j \end{cases}$$

(2) *If $1 \leq n \leq m$, then*

$$n = \prod_{i<m} p_i^{\exp(n,i)}$$

For the proof we need a series of lemmas.

Lemma 8.21.1 [PRA]. $j \neq i \Rightarrow p_j \nmid p_i^n$

Proof. We prove this by induction on n. The base case is clear. By induction hypothesis $p_j \nmid p_i^n$ for some n. Since p_i is prime, we derive that $p_j \nmid p_i$. Hence $p_j \perp p_i^n$ and $p_j \perp p_i$. Corollary 8.14 implies that $p_j \perp p_i^{n+1}$ by Fact 8.17. Thus $p_j \nmid p_i^{n+1}$ and this completes the proof. □

Lemma 8.21.2 [PRA-**schema**]. *Let G be a primitive recursive function such that $G(j) = 0$. Then $p_j \nmid \prod_{i<m} p_i^{G(i)}$.*

Proof of the lemma. We prove this by induction on m. The base case is clear. By induction hypothesis $p_j \nmid \prod_{i<m} p_i^{G(i)}$ for some m. Note that Lemma 8.21.1 shows that $p_j \nmid p_m^{G(m)}$. Hence $p_j \perp \prod_{i<m} p_i^{G(i)}$ and $p_j \perp p_m^{G(m)}$. According to Corollary 8.14

$$p_j \perp \prod_{i<m+1} p_i^{G(i)}$$

Hence $p_j \nmid \prod_{i<m+1} p_i^{G(i)}$. □

Proof of the theorem. We set

$$G(i) = \begin{cases} 0 & \text{if } i = j \\ F(i) & \text{otherwise} \end{cases}$$

By Corollary 5.7 we infer that G is primitive recursive. Easy induction on m shows that

$$\prod_{i<m} p_i^{F(i)} = \begin{cases} \prod_{i<m} p_i^{G(i)} & \text{if } m \leq j \\ p_j^{F(j)} \cdot \prod_{i<m} p_i^{G(i)} & \text{if } j < i \end{cases}$$

Hence

$$\exp\left(\prod_{i<m} p_i^{F(i)}, j\right) = \begin{cases} \exp\left(\prod_{i<m} p_i^{G(i)}, j\right) & \text{if } m \leq j \\ \exp\left(p_j^{F(j)} \cdot \prod_{i<m} p_i^{G(i)}, j\right) & \text{if } j < m \end{cases}$$

Lemma 8.21.2 implies that $p_j \nmid \prod_{i<m} p_i^{G(i)}$ for all m. Thus

$$\exp\left(\prod_{i<m} p_i^{F(i)}, j\right) = \begin{cases} 0 & \text{if } m \leq j \\ F(j) & \text{if } j < m \end{cases}$$

This completes the proof of **(1)**.

Next we prove **(2)**. We prove the assertion by induction on n. The base case is straightforward. Suppose now that $n > 1$. We have two cases.

- n is prime. Then $n = p_j$ for some $j < n$ according to Euclid's Theorem 8.7 and

  $$\exp(n, i) = \begin{cases} 0 & \text{if } i \neq j \\ 1 & \text{if } i = j \end{cases}$$

  Easy induction shows that

  $$j < m \Rightarrow n = \prod_{i<m} p_i^{\exp(n,i)}$$

  Since $j < n$, we infer that

  $$n \leq m \Rightarrow n = \prod_{i<m} p_i^{\exp(n,i)}$$

- There are $1 < k, t < n$ such that $n = k \cdot l$. Pick m no smaller than n. By induction hypothesis

  $$k = \prod_{i<m} p_i^{\exp(k,i)},\ l = \prod_{i<m} p_i^{\exp(l,i)}$$

  According to Fact 6.4 we infer

  $$n = k \cdot l = \prod_{i<m} p_i^{\exp(k,i)+\exp(l,i)}$$

  Next **(1)** implies that $\exp(n, i) = \exp(k, i) + \exp(l, i)$ for every $i < m$. Thus

  $$n = \prod_{i<m} p_i^{\exp(n,i)}$$

This proves **(2)**. □

9. Formalization of sequences and course-of-values recursion

In this section we formalize finite sequences in PRA and prove important theorem on recursion scheme.

Definition 9.1.

$$\mathrm{Seq}(n) \equiv 1 \leq n \wedge \exists_{m<n} \forall_{i<n} \big((i < m \Rightarrow 1 \leq \exp(n,i)) \wedge (m \leq i \Rightarrow \exp(n,i) = 0) \big)$$

If $\mathrm{Seq}(n)$, then n is a *sequence*.

Definition 9.2.

$$\mathrm{lh}(n) = \begin{cases} \mu_{m<n} \big[\forall_{i<n} (m \leq i \Rightarrow \exp(n,i) = 0) \big] & \text{if } \mathrm{Seq}(n) \\ 0 & \text{otherwise} \end{cases}$$

If n is a sequence, then $\mathrm{lh}(n)$ is the *length* of n.

Fact 9.3 [PRA]. *Let n be a sequence. Then the following assertions hold.*

(1) $\mathrm{lh}(n) < n$

(2) $1 < \exp(n,i)$ *for all* $i < \mathrm{lh}(n)$.

(3) $\exp(n,i) = 0$ *for all* $\mathrm{lh}(n) \leq i$.

Proof. We prove **(1)**. Note that $\mathrm{Seq}(n)$ implies $1 \leq n$. Theorem 8.21 implies

$$n = \prod_{i<n} p_i^{\exp(n,i)}$$

Uniqueness assertion of Theorem 8.21 implies that $\exp(n,i) = 0$ for all $n \leq i$. This proves that $\mathrm{lh}(n) < n$.

For the proof of **(2)**, **(3)** note that according to $\mathrm{Seq}(n)$ there exists m such that

$$(i < m \Rightarrow 1 \leq \exp(n,i)) \wedge (m \leq i \Rightarrow \exp(n,i) = 0)$$

for all $i < n$. Then $\mathrm{lh}(n) = m$ and the proof is completed. □

Definition 9.4.

$$n_i = \begin{cases} \exp(n,i) \dot{-} 1 & \text{if } \mathrm{Seq}(n) \\ 0 & \text{otherwise} \end{cases}$$

If n is a sequence and $0 \leq i < \mathrm{lh}(n)$, then n_i is the *i-th element* of n.

Corollary 9.5 [PRA]. $\mathrm{Seq}(n) \Rightarrow \forall_i \, n_i < n$

Proof. Follows immediately from Fact 8.20. □

Fact 9.6 [PRA]. $\mathrm{Seq}(n) \Rightarrow n = \prod_{i<\mathrm{lh}(n)} p_i^{1+n_i}$

Proof. From Fact 9.3 we deduce that

$$\exp(n,i) = \begin{cases} 1 + n_i & \text{if } i < \mathrm{lh}(n) \\ 0 & \text{if } \mathrm{lh}(n) \leq i \end{cases}$$

According to Theorem 8.21 we infer the assertion. □

Corollary 9.7 [PRA]. $(\mathrm{Seq}(n) \wedge \mathrm{Seq}(m)) \Rightarrow \big(n = m \Leftrightarrow \forall_i \, n_i = m_i\big)$

Proof. This is immediate consequence of Fact 9.6. □

Definition 9.8. $\max(n) = \mu_{m<n}\left[\mathrm{Seq}(n) \wedge \forall_{i<\mathrm{lh}(n)}\, n_i \leq m\right]$

Fact 9.9 [PRA]. *The following assertions hold.*

(1) *If n is a sequence, then*
$$n_i \leq \max(n) < n$$
for every $i < \mathrm{lh}(n)$.

(2) *If n is an empty sequence, then* $\max(n) = 0$.

(3) *If n is a sequence of positive length, then there exists $i < \mathrm{lh}(n)$ such that $n_i = \max(n)$*

Proof. Suppose that n is a sequence. Corollary 9.5 implies that
$$n_i \leq n \dot{-} 1$$
for all $i < \mathrm{lh}(n)$. Next **(1)** follows from Theorem 7.2.

The assertion **(2)** is straightforward consequence of **(1)**, since n empty implies that $n = 1$.

Now we prove **(3)**. Suppose that n is a sequence of positive length. Then $\max(n) < n$ and $n_i \leq \max(n)$ for all $i < \mathrm{lh}(n)$. Assume that $n_i < \max(n)$ for all $i < \mathrm{lh}(n)$. Then $n_i \leq \max(n) \dot{-} 1$ for all $i < \mathrm{lh}(n)$. This is a contradiction with Theorem 7.2. Thus there exists $i < \mathrm{lh}(n)$ such that $n_i = \max(n)$. □

Remark 9.10. Suppose that n is a sequence of length k. Then we write n by the juxtaposition notation
$$n_0 ... n_{k\dot{-}1}$$
Next assume that m is another sequence of length l and $n_i = m$ for some $0 \leq i < k$. Then we write n by the following notation with parentheses
$$n_0 ... n_{i\dot{-}1}\, (m_0 ... m_{l\dot{-}1})\, n_{i+1} ... n_{k\dot{-}1}$$

Definition 9.11.
$$n \frown m = \begin{cases} \prod_{i<\mathrm{lh}(n)} p_i^{1+n_i} \cdot \prod_{\mathrm{lh}(n)\leq i<\mathrm{lh}(n)+\mathrm{lh}(m)} p_i^{1+m_{i\dot{-}\mathrm{lh}(n)}} & \text{if } \mathrm{Seq}(n) \text{ and } \mathrm{Seq}(m) \\ 0 & \text{otherwise} \end{cases}$$
If n and m are sequences, then $n \frown m$ is their *concatenation*.

Corollary 9.12 [PRA]. *Let n, m be sequences. Then $n \frown m$ is the unique sequence such that the following assertions hold.*

(1) $\mathrm{lh}(n \frown m) = \mathrm{lh}(n) + \mathrm{lh}(m)$

(2) $(n \frown m)_i = \begin{cases} n_i & \text{if } i < \mathrm{lh}(n) \\ m_i & \text{if } \mathrm{lh}(n) \leq i < \mathrm{lh}(n) + \mathrm{lh}(m) \end{cases}$

Proof. Follows from definition of $n \frown m$ and Fact 9.6. □

Corollary 9.13 [PRA]. *The following assertions hold.*

(1) $1 \frown n = n = n \frown 1$ *for every sequence n.*

(2) $(n \frown m) \frown k = n \frown (m \frown k)$

Proof. This is a consequence of Corollary 9.12. □

Definition 9.14.

$$n_{|i} = \begin{cases} n_0 ... n_{i \dot{-} 1} & \text{if } 1 \leq i < \mathrm{lh}(n) \\ 1 & \text{if } \mathrm{Seq}(n) \text{ and } i = 0 \\ 0 & \text{otherwise} \end{cases}$$

Definition 9.15. $n \triangleright m = \begin{cases} n \frown 2^{1+m} & \text{if } \mathrm{Seq}(n) \\ 0 & \text{otherwise} \end{cases}$

Definition 9.16. Let F be a $(\mathrm{k}+1)$-ary primitive recursive function. We define

$$\tilde{F}(\vec{n}, m) = F(\vec{n}, 0) ... F(\vec{n}, m \dot{-} 1)$$

for every m and every $\vec{n}$. Then $\tilde{F}$ is a *course-of-values* function for F.

Fact 9.17 [PRA-**schema**]. *Let F be a $(\mathrm{k}+1)$-ary primitive recursive function. Then $\tilde{F}(\vec{n}, m)$ is a sequence of length m and*

$$\tilde{F}(\vec{n}, m)_i = F(\vec{n}, i)$$

for all $i < m$ and every $\vec{n}$.

Proof. This is a consequence of Theorem 8.21. □

Proposition 9.18 [PRA-**schema**][course-of-values recursion]. *Let H be a k-ary primitive recursive and let G be a $(\mathrm{k}+2)$-ary primitive recursive function. Then there exists a $(\mathrm{k}+1)$-ary primitive recursive function F such that*

$$F(\vec{n}, 0) = H(\vec{n}),\ F(\vec{n}, m+1) = G\left(\vec{n}, m, \tilde{F}(\vec{n}, m+1)\right)$$

for every $\vec{n}$ and m.

Proof. We define

$$\overline{G}(\vec{n}, m, k) = k \triangleright G(\vec{n}, m, k),\ \overline{H}(\vec{n}) = 2^{1+H(\vec{n})}$$

Then $\overline{G}$ is a $(\mathrm{k}+2)$-ary primitive recursive function and $\overline{H}$ is a k-ary primitive recursive function. Let $\overline{F}$ be a function obtained by primitive recursion from $\overline{G}$ and $\overline{H}$. Then

$$\overline{F}(\vec{n}, 0) = 2^{1+H(\vec{n})},\ \overline{F}(\vec{n}, m+1) = \overline{F}(\vec{n}, m) \triangleright G\left(\vec{n}, m, \overline{F}(\vec{n}, m)\right)$$

for every m and $\vec{n}$. Straightforward induction on m implies that

(1) $\overline{F}(\vec{n}, m)$ is a sequence of length $m+1$ for every $\vec{n}$ and m.

(2) $\overline{F}(\vec{n}, k)_i = \overline{F}(\vec{n}, m)_i$ for all $i \leq k \leq m$.

We define $F(\vec{n}, m) = \overline{F}(\vec{n}, m)_m$. Then **(1)** and **(2)** imply that F is $(\mathrm{k}+1)$-ary primitive recursive function and $\tilde{F}(\vec{n}, m+1) = \overline{F}(\vec{n}, m)$. Thus

$$F(\vec{n}, 0) = H(\vec{n}),\ F(\vec{n}, m+1) = G\left(\vec{n}, m, \tilde{F}(\vec{n}, m+1)\right)$$

for every $\vec{n}$ and m. □

Next we study course-of-values recursion for primitive recursive formulas.

Definition 9.19. Let $\Psi(\vec{n}, m, k)$ be a primitive recursive formula. A primitive recursive formula $\Phi(\vec{n}, m)$ with characteristic function C_Φ such that

$$\Phi(\vec{n}, m) \Leftrightarrow \Psi\left(\vec{n}, m, \widetilde{C_\Phi}(\vec{n}, m)\right)$$

is obtained from $\Psi(\vec{n}, m, k)$ by *course-of-values recursion*.

Remark 9.20. Let $\Psi(\vec{n}, m, k)$ be a primitive recursive formula. Then any two formulas obtained by course-of-values recursion from $\Psi(\vec{n}, m, k)$ are provably equivalent in PRA.

Corollary 9.21 [PRA-**schema**]. *Let $\Psi(\vec{n}, m, k)$ be a primitive recursive formula. Then there exists a formula obtained by course-of-values recursion from $\Psi(\vec{n}, m, k)$.*

Proof. Let C_Ψ be a characteristic function of $\Psi(\vec{n}, m, k)$. Then Proposition 9.18 implies that there exists a primitive recursive function C such that

$$C(\vec{n}, 0) = C_\Psi(\vec{n}, 0, 1)$$

and

$$C(\vec{n}, m+1) = C_\Psi\left(\vec{n}, m+1, \tilde{C}(\vec{n}, m+1)\right)$$

Then $\Phi(\vec{n}, m) \equiv C(\vec{n}, m) = 1$ is a primitive recursive formula that satisfies the assertion. □

10. Formalization of first-order logic in Primitive Recursive Arithmetic

In this section we formalize in PRA our presentation of first-order logic in [Monygham, 2026]. This is arithmetization of first-order meta-theory.

Definition 10.1. We define *logical constants*.

$$\bot = 3, \to = 5, \wedge = 7, \vee = 9, \forall = 11, \exists = 13, = = 15$$

Definition 10.2. A *signature* consists of the following data.

- A primitive recursive formula $\mathrm{Rel}(r)$ and a primitive recursive function arr such that
$$\mathrm{Rel}(r) \Rightarrow (20 < r \wedge \mathrm{rem}(r, 6) = 1)\,, \mathrm{PRA} \vdash \mathrm{Rel}(r) \Rightarrow \mathrm{arr}(r) \neq 0$$
If $\mathrm{Rel}(r)$ holds, then r is a *relation symbol* and $\mathrm{arr}(r)$ is the *arity* of r.
- A primitive recursive formula $\mathrm{Fun}(f)$ and a primitive recursive function arf such that
$$\mathrm{Fun}(f) \Rightarrow (20 < f \wedge \mathrm{rem}(f, 6) = 3)$$
If $\mathrm{Fun}(f)$ holds, then f is a *function symbol* and $\mathrm{arf}(f)$ is the *arity* of f.

Definition 10.3.

$$\mathrm{Var}(x) \equiv (10 < x \wedge \mathrm{rem}(x, 6) = 5)$$

If $\mathrm{Var}(x)$, then x is a *variable*.

Fact 10.4 [PRA]. *At most one of the following assertions hold.*

(1) *n is a sequence.*

(2) *n is a logical constant.*

(3) *n is a relation symbol.*

(4) *n is a function symbol.*

(5) *n is a variable.*

Proof. Follows immediately from definitions and the fact that if n is a sequence, then either $n = 1$ or $2|n$. □

Definition 10.5. We define $\mathrm{Term}(t)$ by course-of-values recursion on the structure of t as follows.

(1) t is a variable.

(2) t is a sequence of the form $ft_1...t_n$ for $n + 1 = \mathrm{lh}(t)$, where f is a n-ary function symbol and $\mathrm{Term}(t_i)$ holds for all $1 \leq i \leq n$.

If $\mathrm{Term}(t)$, then t is a *term*.

Definition 10.6. $\mathrm{Eq}(\phi)$ holds if ϕ is a sequence of the form $t_1 = t_2$, where t_1 and t_2 are terms.

If $\mathrm{Eq}(\phi)$, then ϕ is an *equational formula*.

Definition 10.7. $\mathrm{RelAtomic}(\phi)$ holds if either ϕ is $\bot$ or it is a sequence of the form $rt_1...t_n$ for $n+1 = \mathrm{lh}(\phi)$, where r is n-ary relation symbol and t_i is a term for every $1 \leq i \leq n$.

If $\mathrm{RelAtomic}(\phi)$, then ϕ is an *relational formula*.

Definition 10.8. $\mathrm{Atomic}(\phi)$ holds if either ϕ is relational formula or equational formula.

If $\mathrm{Atomic}(\phi)$, then ϕ is an *atomic formula*.

Remark 10.9. Let x be a variable. Then we write the sequences $\forall x$ and $\exists x$ as $\forall_x$ and $\exists_x$, respectively.

Definition 10.10. Let $\mathrm{Form}(\phi)$ be a primitive recursive formula obtained by course-of-values recursion on the structure on ϕ as follows.

(1) ϕ is an atomic formula.

(2) ϕ is a sequence of one of the forms

$$\phi_1 \wedge \phi_2,\ \phi_1 \vee \phi_2,\ \phi_1 \to \phi_2$$

and $\mathrm{Form}(\phi_1), \mathrm{Form}(\phi_2)$ hold.

(3) ϕ is a sequence of one of the forms

$$\forall_x \psi,\ \exists_x \psi$$

where x is a variable and $\mathrm{Form}(\psi)$ holds.

If $\mathrm{Form}(\phi)$, then ϕ is an *formula*.

Remark 10.11. Let ϕ be a formula. Then we denote $\phi \to \bot$ by $\neg\phi$.

Let ϕ, ψ be formulas. Then we denote $(\phi \to \psi) \wedge (\psi \to \phi)$ by $\phi \leftrightarrow \psi$.

Definition 10.12. Let $\mathrm{Var}(x, n)$ be a primitive recursive formula obtained by course-of-values recursion on the structure of n as follows.

(1) x is a variable and $n = x$.

(2) x is a variable and n is a term of the form $ft_1...t_k$ for $k+1 = \mathrm{lh}(n)$ such that $\mathrm{Var}(x, t_i)$ for some $1 \leq i \leq k$.

(3) x is a variable and n is a relational formula of the form $rt_1...t_k$ for $k+1 = \mathrm{lh}(n)$ such that $\mathrm{Var}(x, t_i)$ for some $1 \leq i \leq k$.

(4) x is a variable and n is an equational formula of the form $t_1 = t_2$ and either $\mathrm{Var}(x, t_1)$ or $\mathrm{Var}(x, t_2)$.

(5) x is a variable and n is a formula of one of the forms

$$\phi_1 \wedge \phi_2,\ \phi_1 \vee \phi_2,\ \phi_1 \to \phi_2$$

and either $\mathrm{Var}(x, \phi_1)$ or $\mathrm{Var}(x, \phi_2)$.

(6) x is a variable and n is a formula of one of the forms

$$\forall_y \psi,\ \exists_y \psi$$

and either $x = y$ or $\mathrm{Var}(x, \psi)$.

If $\mathrm{Var}(x, n)$, then x is a variable in n.

Definition 10.13. Let $\mathrm{Free}(x, \phi)$ be a primitive recursive formula obtained by course-of-values recursion on the structure of ϕ as follows.

(1) x is a variable and ϕ is an atomic formula and $\mathrm{Var}(x, \phi)$.

(2) x is a variable and ϕ is a formula of one of the forms

$$\phi_1 \wedge \phi_2,\ \phi_1 \vee \phi_2,\ \phi_1 \to \phi_2$$

and either $\mathrm{Free}(x, \phi_1)$ or $\mathrm{Free}(x, \phi_2)$.

(3) x is a variable and ϕ is a formula of one of the forms

$$\forall_y \psi,\ \exists_y \psi$$

and $x \neq y$ and $\mathrm{Free}(x, \psi)$.

If $\mathrm{Free}(x, \phi)$, then x is a *free* variable in ϕ.

Fact 10.14 [PRA]. *The following assertions hold.*

(1) $\mathrm{Var}(x, n) \Rightarrow x \leq n$

(2) $\mathrm{Var}(x, n) \Rightarrow (\mathrm{Var}(x) \wedge (\mathrm{Term}(n) \wedge \mathrm{Form}(n)))$

(3) $\mathrm{Free}(x, \phi) \Rightarrow \mathrm{Var}(x, \phi)$

Proof. The proof goes by complete induction schema of Theorem 7.1. The details are left for the reader. □

Fact 10.15 [PRA]. *There exists a primitive recursive function* free *such that* $\mathrm{free}(\phi)$ *is an increasing sequence of all free variables in* ϕ *for every formula* ϕ.

Proof. Left for the reader as an exercise. □

Next we define notion of substitutability.

Definition 10.16. Let $\mathrm{Sub}(t, x, \phi)$ be a primitive recursive formula obtained by course-of-values recursion on the structure of ϕ as follows.

(1) t is a term and x is a variable and ϕ is an atomic formula.

(2) t is a term and x is a variable and ϕ is a formula of one of the forms

$$\phi_1 \wedge \phi_2,\ \phi_1 \vee \phi_2,\ \phi_1 \to \phi_2$$

and $\mathrm{Sub}(t, x, \phi_1)$ and $\mathrm{Sub}(t, x, \phi_2)$.

(3) t is a term and x is a variable and ϕ is a formula of one of the forms

$$\forall_y \psi,\ \exists_y \psi$$

and $x \neq y$ and $\neg\mathrm{Var}(y, t)$ and $\mathrm{Sub}(t, x, \psi)$.

(4) t is a term and x is a variable and ϕ is a formula of one of the forms

$$\forall_x \psi,\ \exists_x \psi$$

If $\mathrm{Sub}(t, x, \phi)$, then t is *substitutable* for x in ϕ.

Definition 10.17. We define primitive recursive substitution function obtained by course-of-values recursion on the structure of first argument as follows.

$$s[x/t] = \begin{cases} t & \text{if } \mathrm{Term}(t) \wedge \mathrm{Var}(x) \wedge s = x \\ s & \text{if } \mathrm{Var}(s) \wedge \mathrm{Var}(x) \wedge s \neq x \\ s_0 s_1[x/t] \ldots s_{\mathrm{lh}(s) \dot{-} 1}[x/t] & \text{if } \mathrm{Term}(t) \wedge \mathrm{Var}(x) \wedge \mathrm{Term}(s) \wedge \mathrm{lh}(s) > 0 \end{cases}$$

$$\phi[x/t] = \begin{cases} \phi_0[x/t] = \phi_2[x/t] & \text{if } \mathrm{Term}(t) \wedge \mathrm{Var}(x) \wedge \mathrm{Eq}(\phi) \\ \phi_0 \phi_1[x/t] \ldots \phi_{\mathrm{lh}(\phi) \dot{-} 1}[x/t] & \text{if } \mathrm{Term}(t) \wedge \mathrm{Var}(x) \wedge \mathrm{RelAtomic}(\phi) \\ \phi_0[x/t] \wedge \phi_2[x/t] & \text{if } \mathrm{Sub}(t, x, \phi) \wedge \phi_1 = \wedge \\ \phi_0[x/t] \vee \phi_2[x/t] & \text{if } \mathrm{Sub}(t, x, \phi) \wedge \phi_1 = \vee \\ \phi_0[x/t] \to \phi_2[x/t] & \text{if } \mathrm{Sub}(t, x, \phi) \wedge \phi_1 = \to \\ \forall_{\phi_1} \phi_2[x/t] & \text{if } \mathrm{Sub}(t, x, \phi) \wedge \phi_0 = \forall \wedge \phi_1 \neq x \\ \exists_{\phi_1} \phi_2[x/t] & \text{if } \mathrm{Sub}(t, x, \phi) \wedge \phi_0 = \exists \wedge \phi_1 \neq x \\ \phi & \text{if } \mathrm{Sub}(t, x, \phi) \wedge \phi_0 = \forall \wedge \phi_1 = x \\ \phi & \text{if } \mathrm{Sub}(t, x, \phi) \wedge \phi_0 = \exists \wedge \phi_1 = x \end{cases}$$

We also set $n[x/t] = 0$ for all other n.

Fact 10.18 [PRA]. *The following assertions hold.*

(1) *If t, s are terms and x is a variable, then $s[x/t]$ is a term.*

(2) *If ϕ is a formula and t is substitutable for x in ϕ, then $\phi[x/t]$ is a formula.*

Proof. The assertion **(1)** follows from definition $s[x/t]$.

The proof of **(2)** goes by induction on ϕ. The details are left for the reader. □

Next we formalize Hilbert-style classical calculus in PRA.

Definition 10.19.

$$\begin{aligned} &\mathrm{PAxiom}(\phi) \equiv \exists_{\phi_1, \phi_2, \phi_3 < \phi} \Big(\mathrm{Form}(\phi_1) \wedge \mathrm{Form}(\phi_2) \wedge \mathrm{Form}(\phi_3) \wedge \Big(\\ &\phi = (\phi_1 \to (\phi_2 \to \phi_3)) \to ((\phi_1 \to \phi_2) \to (\phi_1 \to \phi_3)) \\ &\vee \phi = \phi_1 \to (\phi_2 \to \phi_1) \\ &\vee \phi = (\phi_1 \wedge \phi_2) \to \phi_1 \vee \phi = (\phi_1 \wedge \phi_2) \to \phi_2 \\ &\vee \phi = \phi_1 \to (\phi_2 \to (\phi_1 \wedge \phi_2)) \\ &\vee \phi = \phi_1 \to (\phi_1 \vee \phi_2) \vee \phi = \phi_2 \to (\phi_1 \vee \phi_2) \\ &\vee \phi = (\phi_1 \to \phi_3) \to ((\phi_2 \to \phi_3) \to ((\phi_1 \vee \phi_2) \to \phi_3)) \\ &\vee \phi = ((\phi_1 \to \bot) \to \bot) \to \phi_1 \Big) \Big) \end{aligned}$$

If $\mathrm{PAxiom}(\phi)$, then ϕ is a *propositional axiom*.

Definition 10.20.

$$\begin{aligned}\mathrm{QAxiom}(\phi) \equiv \exists_{\phi_1,\phi_2,\phi_3,x,t<\phi}\Big(&\mathrm{Form}(\phi_1) \wedge \mathrm{Form}(\phi_2) \wedge \mathrm{Var}(x) \wedge \mathrm{Term}(t) \wedge \Big(\\ &\phi = \forall_x(\phi_1 \to \phi_2) \to (\forall_x \phi_1 \to \forall_x \phi_2)\\ &\vee \big(\phi = \phi_1 \to \forall_x \phi_1 \wedge (\neg\mathrm{Free}(x,\phi_1))\big)\\ &\vee \big(\phi = \forall_x \phi_1 \to \phi_1[x/t] \wedge \mathrm{Sub}(t,x,\phi_1)\big)\\ &\vee \big(\phi = \phi_1[x/t] \to \exists_x \phi_1 \wedge \mathrm{Sub}(t,x,\phi_1)\big)\\ &\vee \big(\phi = \forall_x(\phi_1 \to \phi_2) \to (\exists_x \phi_1 \to \phi_2) \wedge (\neg\mathrm{Free}(x,\phi_2))\big)\Big)\Big)\end{aligned}$$

If $\mathrm{QAxiom}(\phi)$, then ϕ is a *quantifier axiom*.

Definition 10.21.

$$\begin{aligned}\mathrm{EqAxiom}(\phi) \equiv \exists_{t_1,t_2,t_3,x,t,\psi<\phi}\Big(&\mathrm{Term}(t_1) \wedge \mathrm{Term}(t_2) \wedge \mathrm{Term}(t_3) \wedge\\ &\mathrm{Var}(x) \wedge \mathrm{Term}(t) \wedge \mathrm{Atomic}(\psi) \wedge \Big(\phi = t_1 = t_1\\ &\vee \phi = t_1 = t_2 \to t_2 = t_1\\ &\vee \phi = t_1 = t_2 \to (t_2 = t_3 \to t_1 = t_3)\\ &\vee \phi = t_1 = t_2 \to t[x/t_1] = t[x/t_2]\\ &\vee \phi = t_1 = t_2 \to (\psi[x/t_1] \to \psi[x/t_2])\Big)\Big)\end{aligned}$$

If $\mathrm{EqAxiom}(\phi)$, then ϕ is an *equational axiom*.

Definition 10.22. Let $\mathrm{Axiom}(\phi)$ be a primitive recursive formula obtained by course-of-values recursion on the structure of ϕ as follows.

(1) ϕ is a propositional axiom.

(2) ϕ is a quantifier axiom.

(3) ϕ is an equational axiom.

(4) ϕ is a formula of the form $\forall_x \psi$ and $\mathrm{Axiom}(\psi)$ holds.

If $\mathrm{Axiom}(\phi)$, then ϕ is a *logical axiom*.

Definition 10.23. $\mathrm{MP}(\phi,\psi,\rho) \equiv \mathrm{Form}(\phi) \wedge \mathrm{Form}(\psi) \wedge \psi = \phi \to \rho$

If $\mathrm{MP}(\phi,\psi,\rho)$, then ρ is obtained by *modus ponens* from ψ and ϕ.

Definition 10.24. Let $\gamma(n)$ be a formula of PRA such that

$$\mathrm{PRA} \vdash \gamma(n) \Rightarrow \mathrm{Form}(n)$$

Then γ is a *predicate on formulas*.

Definition 10.25. Let $\gamma(n)$ be a predicate of formulas. Then we define

$$\mathrm{Proof}_\gamma(p) \equiv \mathrm{lh}(p) > 0 \wedge \forall_{i<\mathrm{lh}(p)}\big(\mathrm{Axiom}(p_i) \vee \gamma(p_i) \vee \exists_{j,k<i} \mathrm{MP}(p_j,p_k,p_i)\big)$$

If $\mathrm{Proof}_\gamma(p)$, then p is a *proof* from γ.

Definition 10.26. Let $\gamma(n)$ be a predicate of formulas. Then we define

$$\mathrm{Prf}_\gamma(p,\phi) \equiv \mathrm{Form}(\phi) \wedge \mathrm{Proof}_\gamma(p) \wedge p[\mathrm{lh}(p) \dot{-} 1] = \phi$$

If $\mathrm{Prf}_\gamma(p,\phi)$, then p is a *proof* of ϕ from γ.

Fact 10.27 [PRA-**schema**]. *Let $\gamma(n)$ be a primitive recursive predicate of formulas. Then* $\mathrm{Proof}_\gamma(p)$ *and* $\mathrm{Prf}_\gamma(p,\phi)$ *are primitive recursive.*

Proof. Follows immediately from the definition. □

Definition 10.28. Let $\gamma(n)$ be a predicate of formulas. Then we define

$$\mathrm{Pr}_\gamma(\phi) \equiv \exists_p \mathrm{Prf}_\gamma(p,\phi)$$

The formula $\mathrm{Pr}_\gamma(\phi)$ is the *Gödel provability predicate* for γ.

11. Arithmetization of primitive recursive arithmetic

We formalize PRA in itself according to the method presented in the previous section. We begin by describing the language **PRA** of primitive recursive arithmetic. We adopt the notational convention that

$$n_0...n_k = [n_0,...,n_k]$$

for every sequence n. We use this notation simultaneously with the juxtaposition notation of Remark 9.10. This is convenient in the context of encoding.

Definition 11.1.

$$Z(f) \equiv \exists_{k<f} f = 6\cdot[0,k]+3$$

If $Z(f)$, then f is a *zero function symbol*.

Definition 11.2. $s = 6\cdot[1,1]+3$ is the *successor function symbol*.

Definition 11.3.

$$I(f) \equiv \exists_{k<f}\exists_{i\leq k}\,(k\neq 0\wedge i\neq 0\wedge f = 6\cdot[2,k,i]+3)$$

If $I(f)$, then f is a *projection function symbol*.

Definition 11.4. We set

$$\mathrm{arf}(f) = \mathrm{div}(f,6)_1$$

We define $\mathrm{PrimFun}(f)$ formula by course-of-values recursion on the structure of f as follows.

(1) f is a zero function symbol.

(2) f is the succesor function symbol.

(3) f is a projection function symbol.

(4) There exist $m,g,f_1...f_k < f$ such that $f = 6\cdot[3,m,g,f_1...f_k]+3$ and $\mathrm{PrimFun}(g)$ and $\mathrm{arf}(g)=k$ and $\mathrm{PrimFun}(f_i)$ and $\mathrm{arf}(f_i)=m$ for every $1\leq i\leq k$.

(5) There exist $k,g,h<f$ such that $f = 6\cdot[4,k+1,g,h]+3$ and $\mathrm{PrimFun}(g)$ and $\mathrm{arf}(g) = k+2$ and $\mathrm{PrimFun}(h)$ and $\mathrm{arf}(h)=k$.

If $\mathrm{PrimFun}(f)$, then f is a *primitive recursive function symbol* and $\mathrm{arf}(f)$ is its *arity*.

Definition 11.5. The *language of primitive recursive arithmetic* is the first-order language with signature that consists of no relation symbols and primitive recursive function symbols. We denote this language by **PRA**.

Definition 11.6.

$$\mathrm{VarSeq}(\vec{x}) \equiv \forall_{i<\mathrm{lh}(\vec{x})}\,\mathrm{Var}(\vec{x}_i)$$

If $\mathrm{VarSeq}(\vec{x})$, then $\vec{x}$ is a *variable sequence*.

Remark 11.7. We introduce notational conventions.

We denote $6\cdot[0,0]+3$ by o. Note that o is a constant symbol.

We denote $6 \cdot [0, k] + 3$ by z_k.

We denote $6 \cdot [2, k, i] + 3$ for $1 \leq i \leq k$ by ι^i_k.

Suppose that f is a primitive recursive function symbol such that

$$f = 6 \cdot [3, m, g, f_1 ... f_k] + 3$$

We denote f by $\mathrm{comp}(g, f_1 ... f_k)$. Note that for f of this form $g, f_1, ..., f_k$ values of primitive recursive functions of f.

Suppose that f is a primitive recursive function symbol such that

$$f = 6 \cdot [4, k+1, g, h] + 3$$

We denote f by $\mathrm{prim}(g, h)$. Note that for f of this form g, h are values of primitive recursive functions of f.

Suppose that $\vec{x}$ is a variable sequence and y is a variable, then we denote by $\vec{x}y$ the sequence obtained by appending $\vec{x}$ with y. Similarly, $\vec{x}s(y)$ and $\vec{x}o$ denote the sequences obtained by appending $\vec{x}$ with sy and o, respectively. Note that this two sequences are not variable sequences.

If f is a primitive recursive function symbol and $\vec{x}$ is a variable sequence such that the length of $\vec{x}$ coincides with the arity of f, then we write $f\vec{x}$. Similarly, we write $f\vec{x}sy$ or $f\vec{x}o$, if the length of $\vec{x}sy$ and $\vec{x}o$ coincides with arity of f.

If $\vec{x} = x_1 ... x_k$ is a sequence of variables, then we denote sequence $\forall_{x_1} ... \forall_{x_k}$ by $\forall_{\vec{x}}$.

12. Gödel coding and fixed point meta-lemma

This section has meta-theoretic character. Prior to establishing Gödel coding we need to make the presentation of the syntax in Section 3 more precise. Since primitive recursive functions in Section 3 are rigorously described and there are no relation symbols, the missing part is specification of variables of PRA. For this we use meta-level numbers and assume that each variable of PRA corresponds to meta-level number n. We denote this correspondence by $\mathrm{n} \mapsto \mathrm{n}_\mathrm{n}$.

Definition 12.1. For each primitive recursive function F we assign a function symbol $\ulcorner F \urcorner$ of **PRA** by induction on the construction of F.

(1) $\ulcorner \mathbf{Z}_\mathrm{k} \urcorner = 6 \cdot [0, \underbrace{\mathbf{S} ... \mathbf{S}}_{\mathrm{k}}(0)] + 3$ for every meta-level number k.

(2) $\ulcorner \mathbf{S} \urcorner = 6 \cdot [1, 1] + 3$

(3) $\ulcorner \mathbf{I}^\mathrm{i}_\mathrm{k} \urcorner = 6 \cdot [2, \underbrace{\mathbf{S} ... \mathbf{S}}_{\mathrm{k}}(0), \underbrace{\mathbf{S} ... \mathbf{S}}_{\mathrm{i}}(0)] + 3$ for every $1 \leq \mathrm{i} \leq \mathrm{k}$.

(4) $\ulcorner \mathrm{comp}(G, F_1, ..., F_\mathrm{k} \urcorner = 6 \cdot [3, \underbrace{\mathbf{S} ... \mathbf{S}}_{\mathrm{m}}(0), \ulcorner G \urcorner, \ulcorner F_1 \urcorner ... \ulcorner F_\mathrm{k} \urcorner] + 3$ where G is k-ary primitive recursive function and $F_1, ..., F_\mathrm{k}$ are m-ary primitive recursive functions.

(5) $\ulcorner \mathrm{prim}(G, H) \urcorner = 6 \cdot [4, \underbrace{\mathbf{S} ... \mathbf{S}}_{\mathrm{k+1}}(0), \ulcorner G \urcorner, \ulcorner H \urcorner] + 3$ where G is $\mathrm{k}+2$-ary primitive recursive function and H is k-ary primitive recursive function.

The function symbol $\ulcorner F \urcorner$ is the *Gödel code* of F.

Next we extend Gödel coding to terms and formulas of PRA.

Definition 12.2. For each term and formula of PRA we assign an expression of **PRA** by induction as follows.

(1) We set $\ulcorner n_{\mathrm{n}} \urcorner = 6 \cdot \underbrace{\mathbf{S}...\mathbf{S}}_{\mathrm{n}}(0) + 11$ for meta-level number n.

(2) We set $\ulcorner Ft_1...t_{\mathrm{k}} \urcorner = \ulcorner F \urcorner \ulcorner t_1 \urcorner ... \ulcorner t_{\mathrm{k}} \urcorner$ for every primitive recursive function F of arity k and terms $t_1, ..., t_{\mathrm{k}}$ of PRA.

(3) We set

$$\ulcorner t_1 = t_2 \urcorner = \ulcorner t_1 \urcorner = \ulcorner t_2 \urcorner,$$
$$\ulcorner \Phi(\vec{n}) \Rightarrow \Psi(\vec{n}) \urcorner = \ulcorner \Phi(\vec{n}) \urcorner \to \ulcorner \Psi(\vec{n}) \urcorner, \ulcorner \Phi(\vec{n}) \wedge \Psi(\vec{n}) \urcorner = \ulcorner \Phi(\vec{n}) \urcorner \wedge \ulcorner \Psi(\vec{n}) \urcorner,$$
$$\ulcorner \Phi(\vec{n}) \vee \Psi(\vec{n}) \urcorner = \ulcorner \Phi(\vec{n}) \urcorner \vee \ulcorner \Psi(\vec{n}) \urcorner, \ulcorner \neg \Phi(\vec{n}) \urcorner = \ulcorner \Phi(\vec{n}) \urcorner \to \bot$$
$$\ulcorner \forall_n \Phi(\vec{n}) \urcorner = \forall_{\ulcorner n \urcorner} \ulcorner \Phi(\vec{n}) \urcorner, \ulcorner \exists_n \Phi(\vec{n}) \urcorner = \exists_{\ulcorner n \urcorner} \ulcorner \Phi(\vec{n}) \urcorner$$

for every terms t_1, t_2, every formulas $\Phi(\vec{n}), \Psi(\vec{n})$ and every variable n of PRA.

Fact 12.3 [PRA-**schema**]. *Let e be an expression of* PRA. *Then there exists meta-level number* n *such that*

$$\ulcorner e \urcorner = \underbrace{\mathbf{S}...\mathbf{S}}_{\mathrm{n}}(0)$$

Proof. Left for the reader as an exercise. □

Remark 12.4. We want to explain application of the diagonalization method discovered by Cantor and applied by Gödel in his seminal work [Gödel, 1931]. Suppose that O is a type of objects, S is a type of sentences and P is a type of predicates i.e. functions $O \to S$. Assume that there is a function

$$\mathrm{name} : P \coprod S \to O$$

that sends predicate or sentence to its *name* and an *evaluator* function

$$\mathrm{eval} : O \times O \to O$$

such that

$$\mathrm{eval}\big(\mathrm{name}(p), o\big) = \mathrm{name}(p(o))$$

for every p in P and o in O. Define $\mathrm{diag} : O \to O$ by formula

$$\mathrm{diag}(o) = \mathrm{eval}(o, o)$$

and assume that the composition $p \cdot \mathrm{diag}$ is in P for every p in P. Then we claim that every p in P admits a fixed point i.e. there exists s_p in S such that $p\left(\mathrm{name}(s_p)\right) = s_p$. The proof is fairly simple. Fix p and consider $q : O \to S$ such that

$$q(o) = p\left(\mathrm{diag}(o)\right)$$

Then q is in P. Let $s_p = q\left(\mathrm{name}(q)\right)$. Then

$$s_p = q\big(\mathrm{name}(q)\big) = p\Big(\mathrm{diag}\big(\mathrm{name}(q)\big)\Big) =$$

$$= p\Big(\mathrm{eval}\big(\mathrm{name}(q), \mathrm{name}(q)\big)\Big) = p\Big(\mathrm{name}\big(q(\mathrm{name}(q))\big)\Big) = p\big(\mathrm{name}(s_p)\big)$$

Definition 12.5.

$$\overline{n+1} = s\overline{n}, \overline{0} = o$$

The term $\overline{n}$ of **PRA** is a *numeral*.

Now we apply the idea described in Remark 12.4 to Gödel coding defined above.

Theorem 12.6 [PRA-**schema**][Fixed point lemma]. *For every formula $\Phi(n)$ of* PRA *there exists a sentence F_Φ such that*

$$\Phi(\ulcorner F_\Phi \urcorner) \Leftrightarrow F_\Phi$$

Proof. Without loss of generality assume that n corresponds to n_1 and $x = 6 \cdot \mathbf{S}(0) + 11 = 17$. Clearly x is a variable of **PRA**. We define

$$\text{eval}(\phi, n) = \begin{cases} \phi[x/\overline{n}] & \text{if Form}(\phi) \wedge \text{Free}(x, \phi) \\ 0 & \text{otherwise} \end{cases}$$

Consider $\Psi(n) = \Phi(\text{diag}(n))$ where $\text{diag}(n) = \text{eval}(n, n)$. Let $\psi = \ulcorner \Psi(n) \urcorner$. Then ψ is a formula of **PRA** having x as a free variable and according to Fact 12.3 there exists meta-level number n such that

$$\psi = \underbrace{\mathbf{S}...\mathbf{S}}_{\text{n}}(0)$$

We set $F_\Phi \equiv \Psi(\underbrace{\mathbf{S}...\mathbf{S}}_{\text{n}}(0))$. Note that

$$\ulcorner F_\Phi \urcorner = \ulcorner \Psi(\underbrace{\mathbf{S}...\mathbf{S}}_{\text{n}}(0)) \urcorner = \psi[x/\underbrace{s...s}_{\text{n}}\, o] = \text{eval}(\psi, \underbrace{\mathbf{S}...\mathbf{S}}_{\text{n}}(0))$$

Then

$$F_\Phi \equiv \Psi(\underbrace{\mathbf{S}...\mathbf{S}}_{\text{n}}(0)) \Leftrightarrow \Psi(\psi) \equiv \Phi(\text{diag}(\psi)) \equiv$$

$$\equiv \Phi\left(\text{eval}(\psi, \psi)\right) \Leftrightarrow \Phi(\text{eval}(\psi, \underbrace{\mathbf{S}...\mathbf{S}}_{\text{n}}(0)) \equiv \Phi(\ulcorner F_\Phi \urcorner)$$

This completes the proof. □

13. Representability of primitive recursive functions

In this section we prove important result concerning primitive recursive parametrizations of proofs of equalities in PRA.

Remark 13.1. Suppose that $\vec{n} = (n_1, ..., n_\text{k})$ for some meta-level k. Then we set

$$\vec{\overline{n}} = \overline{n_1}...\overline{n_\text{k}}$$

Definition 13.2. Let Γ be a collection of formulas of the language of PRA. Suppose that Γ contains all axioms of PRA. Then Γ is an *extension* of PRA.

Theorem 13.3 [**Γ-schema**]. *Let Γ be an extension of* PRA *and let γ be a predicate on formulas. Suppose that $\Gamma \vdash \gamma(\ulcorner \Lambda \urcorner)$ holds for every axiom Λ of* PRA. *Let F be a* k*-ary primitive recursive function. There exists a primitive recursive function D_F such that*

$$\Gamma \vdash \forall_{\vec{n}} \forall_m \left(F(\vec{n}) = m \Rightarrow \text{Prf}_\gamma \left(D_F(\vec{n}), \ulcorner F \urcorner \vec{\overline{n}} = \overline{m} \right) \right)$$

where $\vec{n} = (n_1, ..., n_\text{k})$.

Proof. We construct D_F using meta-theoretic course-of-values recursion on the complexity of F. Note that for fixed F this gives effective procedure that constructing D_F.

(1) Fix some meta-level number k. We define $D_{\mathbf{Z}_\text{k}}$ to be a primitive recursive function such that $D_{\mathbf{Z}_\text{k}}(\vec{n})$ is a proof of

$$\ulcorner \mathbf{Z}_\text{k} \urcorner \vec{\overline{n}} = \overline{\mathbf{Z}_\text{k}(\vec{n})}$$

from γ obtained as an application of universal instantation with $\vec{\overline{n}}$ to the axiom

$$\forall_{\vec{x}} \ulcorner \mathbf{Z}_\text{k} \urcorner \vec{x} = o$$

(2) We define $D_\mathbf{S}$ to be a primitive recursive function such that $D_\mathbf{S}(n)$ is a proof of

$$s\overline{n} = \overline{\mathbf{S}(n)}$$

from γ obtained as application of the reflexivity axiom.

(3) Fix $1 \leq \mathrm{i} \leq \mathrm{k}$. We define $D_{\mathbf{I}^{\mathrm{i}}_{\mathrm{k}}}$ to be a primitive recursive function such that $D_{\mathbf{I}^{\mathrm{i}}_{\mathrm{k}}}(\vec{n})$ is a proof of

$$\ulcorner \mathbf{I}^{\mathrm{i}}_{\mathrm{k}} \urcorner \vec{\overline{n}} = \overline{\mathbf{I}^{\mathrm{i}}_{\mathrm{k}}(\vec{n})}$$

from γ obtained as an application of universal instantiation with $\vec{\overline{n}}$ to the axiom

$$\forall_{\vec{x}} \ulcorner \mathbf{I}^{\mathrm{i}}_{\mathrm{k}} \urcorner \vec{x} = x_i$$

where $i = \underbrace{\mathbf{S}...\mathbf{S}}_{\mathrm{i}}(0)$.

(4) Suppose that F is $\mathrm{comp}(G, F_1, ..., F_{\mathrm{k}})$ for some k-ary primitive recursive G and m-ary primitive recursive F_{i} for $1 \leq \mathrm{i} \leq \mathrm{k}$. Moreover, assume by meta-theoretic recursion that D_G and $D_{F_{\mathrm{i}}}$ for $1 \leq \mathrm{i} \leq \mathrm{k}$ are constructed.

Let P be a primitive recursive function given by

$$P(\vec{n}) = D_{F_1}(\vec{n}) \frown ... \frown D_{F_{\mathrm{k}}}(\vec{n}) \frown D_G(\vec{m})$$

where $\vec{m} = (F_1(\vec{n}), ..., F_{\mathrm{k}}(\vec{n}))$. Next let Q be pimitive recursive function such that $Q(\vec{n})$ is a proof of

$$\ulcorner F \urcorner \vec{\overline{n}} = \overline{F(\vec{n})}$$

from

$$\ulcorner G \urcorner \vec{\overline{m}} = \overline{G(\vec{m})}, \ulcorner F_{\mathrm{i}} \urcorner \vec{\overline{n}} = \overline{F_{\mathrm{i}}(\vec{n})}$$

for $1 \leq \mathrm{i} \leq \mathrm{k}$ and γ that uses universal instantiation with $\vec{\overline{n}}$ of the axiom

$$\forall_{\vec{x}} \ulcorner F \urcorner (\vec{x}) = \ulcorner G \urcorner (\ulcorner F_1 \urcorner \vec{x}) ... (\ulcorner F_{\mathrm{k}} \urcorner \vec{x}))$$

combined with equality axioms and observation that $G(\vec{m}) = F(\vec{n})$.

Finally, let D_F be a primitive recursive function defined by formula

$$D_F(\vec{n}) = P(\vec{n}) \frown Q(\vec{n})$$

(5) Suppose that F is $\mathrm{prim}(G, H)$ for some $(\mathrm{k}+2)$-ary primitive recursive function G and k-ary primitive recursive function H. Moreover, assume by meta-theoretic recursion that D_G and D_H are constructed.

Let P be a primitive recursive function such that $P(\vec{n})$ is a proof of

$$\ulcorner F \urcorner \vec{\overline{n}} \overline{o} = \overline{F(\vec{n})}$$

from $\ulcorner H \urcorner \vec{\overline{n}} = \overline{H(\vec{n})}$ and γ that uses universal instantiation of the axiom

$$\forall_{\vec{x}} \ulcorner F \urcorner \vec{x} o = \ulcorner H \urcorner \vec{x}$$

with $\vec{\overline{n}}$ combined with equality axioms and observation that $F(\vec{n}, 0) = H(\vec{n})$.

Next let Q be a primitive recursive function such that $Q(\vec{n}, m)$ is a proof of

$$\ulcorner F \urcorner \vec{\overline{n}} s \overline{m} = \overline{F(\vec{n}, \mathbf{S}(m))}$$

from

$$\ulcorner G \urcorner \vec{\overline{n}} \overline{m} \overline{F(\vec{n}, m)} = \overline{G(\vec{n}, m, F(\vec{n}, m))}, \ulcorner F \urcorner \vec{\overline{n}} \overline{m} = \overline{F(\vec{n}, m)}$$

and γ that uses universal instantiation of the axiom

$$\forall_{\vec{x}} \forall_y \ulcorner F \urcorner \vec{x} s y = \ulcorner G \urcorner \vec{x} y (\ulcorner F \urcorner \vec{x} y)$$

with $\vec{\overline{n}} \overline{m}$ combined with equality axioms and observation that

$$F(\vec{n}, \mathbf{S}(m)) = G(\vec{n}, m, F(\vec{n}, m))$$

Then D_F is defined by primitive recursion by the following formulas

$$D_F(\vec{n}, 0) = D_H(\vec{n}) \frown P(\vec{n})$$

and

$$D_F\left(\vec{n},\mathbf{S}(m)\right)=D_F\left(\vec{n},m\right)\frown D_G\left(\vec{n},m,F(\vec{n},m)\right)\frown Q(\vec{n},m)$$

We left for the reader to verify that D_F satisfies the assertion under the assumptions of the theorem. □

Remark 13.4. Then there is no primitive recursive function D such that

$$\mathrm{PRA}\vdash\forall_n\forall_m\left(F(n)=m\Rightarrow\mathrm{Prf}_\gamma\left(D\left(\ulcorner F\urcorner,n\right),\ulcorner F\urcorner\overline{n}=\overline{m}\right)\right)$$

for every unary primitive recursive functions F. Hence Theorem 13.3 requires separate D_F for each F. In order to see why this is the case first assume that such uniform D exists. The last element of $D\left(\ulcorner F\urcorner,n\right)$ is the formula

$$\ulcorner F\urcorner\overline{n}=\overline{m}$$

Since m can be extracted from $\overline{m}$, we infer that there exists primitive recursive procedure extracting $m=F(n)$ from $D\left(\ulcorner F\urcorner,n\right)$. Thus there exists a universal primitive recursive function i.e. a primitive recursive function U such that

$$U(\ulcorner F\urcorner,n)=F(n)$$

for every primitive recursive unary function F and every n. Let Δ be a primitive recursive function given by formula

$$\Delta(n)=U(n,n)+1$$

Then

$$U(\ulcorner\Delta\urcorner,\ulcorner\Delta\urcorner)=\Delta(\ulcorner\Delta\urcorner)=U(\ulcorner\Delta\urcorner,\ulcorner\Delta\urcorner)+1$$

and hence $0=1$. This is a contradiction.

Proposition 13.5 [Metatheory]. *Let Γ be a collection of formulas of the language of* PRA *and let γ be a predicate of formulas. Then the following assertions hold.*

- **(1)** *Suppose that $\Gamma\vdash\gamma\left(\ulcorner\Phi(\vec{n})\urcorner\right)$ for every formula $\Phi(\vec{n})$ in Γ. If $\Gamma\vdash\Phi(\vec{n})$ for some formula $\Phi(\vec{n})$, then $\Gamma\vdash\mathrm{Pr}_\gamma\left(\ulcorner\Phi(\vec{n})\urcorner\right)$.*
- **(2)** *If Γ is an extension of* PRA*, then*

$$\Gamma\vdash\left(\mathrm{Pr}_\gamma\left(\ulcorner\Phi(\vec{n})\urcorner\right)\wedge\mathrm{Pr}_\gamma\left(\ulcorner\Phi(\vec{n})\Rightarrow\Psi(\vec{n})\urcorner\right)\right)\Rightarrow\mathrm{Pr}_\gamma\left(\ulcorner\Psi(\vec{n})\urcorner\right)$$

for all formulas $\Phi(\vec{n})$ and $\Psi(\vec{n})$.

Proof. Suppose that $\Phi(\vec{n})$ is provable in Γ. According to [Monygham, 2026] a proof of $\Phi(\vec{n})$ is a finite meta-sequence

$$D_1,...,D_{\mathrm{m}}$$

of formulas. Let $d=\ulcorner D_1\urcorner...\ulcorner D_{\mathrm{m}}\urcorner$. Since $\Gamma\vdash\gamma\left(\ulcorner\Psi(\vec{n})\urcorner\right)$ for every $\Psi(\vec{n})$ in Γ, we infer that $\Gamma\vdash\mathrm{Prf}_\gamma\left(d,\ulcorner\Phi(\vec{n})\urcorner\right)$. Hence also $\Gamma\vdash\mathrm{Pr}_\gamma\left(\ulcorner\Phi(\vec{n})\urcorner\right)$. This proves **(1)**.

Let ϕ,ψ be formulas of **PRA**. We have

$$\mathrm{PRA}\vdash\left(\mathrm{Prf}_\gamma(p,\phi)\wedge\mathrm{Prf}_\gamma(q,\phi\to\psi)\right)\Rightarrow\mathrm{Prf}_\gamma\left(p\frown q\lhd\psi,\psi\right)$$

Thus $\mathrm{PRA}\vdash\left(\mathrm{Pr}_\gamma\left(\phi\right)\wedge\mathrm{Pr}_\gamma\left(\phi\to\psi\right)\right)\Rightarrow\mathrm{Pr}_\gamma\left(\psi\right)$. Since Γ is an extension of PRA, we derive that

$$\mathrm{PRA}\vdash\left(\mathrm{Pr}_\gamma\left(\phi\right)\wedge\mathrm{Pr}_\gamma\left(\phi\to\psi\right)\right)\Rightarrow\mathrm{Pr}_\gamma\left(\psi\right)$$

Next we apply this fact to $\phi=\ulcorner\Phi(\vec{n})\urcorner,\psi=\ulcorner\Psi(\vec{n})\urcorner$ and use that

$$\ulcorner\Phi(\vec{n})\Rightarrow\Psi(\vec{n})\urcorner=\ulcorner\Phi(\vec{n})\urcorner\to\ulcorner\Psi(\vec{n})\urcorner$$

to derive **(2)**. □

Definition 13.6. Let $\Phi(\vec{n})$ be a formula such that

$$\Phi(\vec{n}) \equiv \exists_{\vec{m}} \Theta(\vec{m}, \vec{n})$$

for some primitive recursive formula $\Theta(\vec{m}, \vec{n})$. Then $\Phi(\vec{n})$ is a Σ_1-formula.

Theorem 13.7 [Metatheory][Σ_1-completeness]. *Let Γ be a theory extending* PRA *and let γ be a predicate on formulas. Suppose that $\Gamma \vdash \gamma(\ulcorner \Lambda \urcorner)$ for every sentence Λ in Γ. Then*

$$\Gamma \vdash \Xi \Rightarrow \mathrm{Pr}_\gamma\left(\ulcorner \Xi \urcorner\right)$$

for every Σ_1-sentence Ξ.

Proof. Suppose that $\Xi \equiv \exists_{\vec{n}} \Theta(\vec{n})$ where $\Theta(\vec{n})$ is primitive recursive. Let C be a characteristic function of $\Theta(\vec{n})$. Then $\Theta(\vec{n})$ is equivalent with $C(\vec{n}) = 1$ in PRA. Let $\vec{x}$ be a variable sequence of Gödel codes of $\vec{n}$.

Since Γ is an extension of PRA we have

$$\Gamma \vdash \Xi \Leftrightarrow \exists_{\vec{n}} C(\vec{n}) = 1$$

and by Proposition 13.5 we infer

$$\Gamma \vdash \mathrm{Pr}_\gamma(\ulcorner \Xi \urcorner) \Leftrightarrow \mathrm{Pr}_\gamma\left(\exists_{\vec{x}} \ulcorner C \urcorner \vec{x} = \overline{1}\right)$$

On the other hand note that

$$\Gamma \vdash C(\vec{n}) = 1 \Rightarrow \mathrm{Prf}_\gamma\left(q, \ulcorner C \urcorner \vec{\overline{n}} = \overline{1}\right)$$

by Theorem 13.3 for some q and hence

$$\Gamma \vdash C(\vec{n}) = 1 \Rightarrow \mathrm{Pr}_\gamma\left(\ulcorner C \urcorner \vec{\overline{n}} = \overline{1}\right)$$

Existential instantiation implies that

$$\Gamma \vdash \mathrm{Pr}_\gamma\left(\ulcorner C \urcorner \vec{\overline{n}} = \overline{1} \rightarrow \exists_{\vec{x}} \ulcorner C \urcorner \vec{x} = \overline{1}\right)$$

Thus by combining the second assertion in Proposition 13.5 and deduction theorem we infer that

$$\Gamma, C(\vec{n}) = 1 \vdash \mathrm{Pr}_\gamma\left(\exists_{\vec{x}} \ulcorner C \urcorner \vec{x} = \overline{1}\right)$$

Therefore, we have

$$\Gamma \vdash \exists_{\vec{n}} C(\vec{n}) = 1 \Rightarrow \mathrm{Pr}_\gamma\left(\exists_{\vec{x}} \ulcorner C \urcorner \vec{x} = \overline{1}\right)$$

We proved that

1. $\Gamma \vdash \Xi \Leftrightarrow \exists_{\vec{n}} C(\vec{n}) = 1$
2. $\Gamma \vdash \mathrm{Pr}_\gamma(\ulcorner \Xi \urcorner) \Leftrightarrow \mathrm{Pr}_\gamma\left(\exists_{\vec{x}} \ulcorner C \urcorner \vec{x} = \overline{1}\right)$
3. $\Gamma \vdash \exists_{\vec{n}} C(\vec{n}) = 1 \Rightarrow \mathrm{Pr}_\gamma\left(\exists_{\vec{x}} \ulcorner C \urcorner \vec{x} = \overline{1}\right)$

Combining them we derive

$$\Gamma \vdash \Xi \Rightarrow \mathrm{Pr}_\gamma\left(\ulcorner \Xi \urcorner\right)$$

This completes the proof. □

14. Hilbert-Bernays conditions

In this section we discuss formalization of mathematical theories extending PRA. We start by introducing some notions.

Definition 14.1. Let Γ be an collection of formulas of the language of PRA and let γ be a predicate of formulas. The assertions

HB1. Let $\Phi(\vec{n})$ be a formula. If $\Gamma \vdash \Phi(\vec{n})$, then $\Gamma \vdash \mathrm{Pr}_\gamma(\ulcorner \Phi(\vec{n}) \urcorner)$.

HB2. $\Gamma \vdash \big(\mathrm{Pr}_\gamma(\ulcorner \Phi(\vec{n}) \urcorner) \wedge \mathrm{Pr}_\gamma(\ulcorner \Phi(\vec{n}) \Rightarrow \Psi(\vec{n}) \urcorner)\big) \Rightarrow \mathrm{Pr}_\gamma(\ulcorner \Psi(\vec{n}) \urcorner)$ for all formulas $\Phi(\vec{n}), \Psi(\vec{n})$

HB3. $\Gamma \vdash \mathrm{Pr}_\gamma(\ulcorner \Phi(\vec{n}) \urcorner) \Rightarrow \mathrm{Pr}_\gamma\Big(\ulcorner \mathrm{Pr}_\gamma(\ulcorner \Phi(\vec{n}) \urcorner) \urcorner\Big)$ for every formula $\Phi(\vec{n})$.

are called *Hilbert-Bernays conditions* for Γ and γ.

Corollary 14.2 [Metatheory]. *Let Γ be a collection of formulas of the language of* PRA *and let γ be a predicate of formulas. Then the following assertions hold.*

(1) *Suppose that $\Gamma \vdash \gamma(\ulcorner \Phi(\vec{n}) \urcorner)$ for every formula $\Phi(\vec{n})$ in Γ. Then* **HB1** *holds.*

(2) *If Γ is an extension of* PRA*, then* **HB2** *holds.*

(3) *Suppose that the following assertions hold.*

Γ *is a theory extending* PRA.

$\Gamma \vdash \gamma(\ulcorner \Lambda \urcorner)$ *for every sentence Λ in Γ.*

There exists Σ_1-formula $\Psi(n)$ such that

$$\Gamma \vdash \mathrm{Pr}_\gamma(\phi) \Leftrightarrow \Psi(\phi)$$

Then **HB3** *holds.*

Proof. Assertions **(1)** and **(2)** are proved in Proposition 13.5.

We prove **(3)**. Fix a formula $\Phi(\vec{n})$. Then $\Psi(\ulcorner \Phi(\vec{n}) \urcorner)$ is a Σ_1-sentence and

$$\Gamma \vdash \Psi(\ulcorner \Phi(\vec{n}) \urcorner) \Rightarrow \mathrm{Pr}_\gamma\Big(\ulcorner \Psi(\ulcorner \Phi(\vec{n}) \urcorner) \urcorner\Big)$$

by Theorem 13.7. Next by **(1)** and **(2)** we infer that

$$\Gamma \vdash \mathrm{Pr}_\gamma\big(\ulcorner \Psi(\ulcorner \Phi(\vec{n}) \urcorner) \urcorner\big) \Leftrightarrow \mathrm{Pr}_\gamma\Big(\ulcorner \mathrm{Pr}_\gamma(\ulcorner \Phi(\vec{n}) \urcorner) \urcorner\Big)$$

Combining these two we derive that

$$\Gamma \vdash \mathrm{Pr}_\gamma(\ulcorner \Phi(\vec{n}) \urcorner) \Rightarrow \mathrm{Pr}_\gamma\Big(\ulcorner \mathrm{Pr}_\gamma(\ulcorner \Phi(\vec{n}) \urcorner) \urcorner\Big)$$

by Theorem 13.7. This completes the proof. □

Remark 14.3. Note that if γ is primitive recursive, then $\mathrm{Pr}_\gamma(\phi)$ is Σ_1-formula by Fact 10.27.

Moreover, the extension of PRA obtained by allowing induction schema for all Σ_1-formulas proves that $\mathrm{Pr}_\gamma(\phi)$ is equivalent Σ_1-formula if γ is Σ_1-formula.

15. Tarski theorem on undefinability of truth and first incompleteness theorem

In this section we apply fixed point lemma to prove syntactic variant of important result due to Tarski.

Theorem 15.1 [Metatheory][Undefinability of truth]. *Let* Γ *be an extension of* PRA. *If* Γ *is consistent, then there is no formula* $\mathrm{Tr}(n)$ *such that*

$$\Gamma \vdash \Xi \Leftrightarrow \mathrm{Tr}(\ulcorner \Xi \urcorner)$$

for every sentence Ξ.

Proof. Suppose that $\mathrm{Tr}(n)$ exists. Theorem 12.6 implies that there exists a sentence L such that

$$\mathrm{PRA} \vdash \mathrm{L} \Leftrightarrow \neg \mathrm{Tr}(\ulcorner \mathrm{L} \urcorner)$$

Since Γ contains axioms of PRA, we infer that

$$\Gamma \vdash \mathrm{L} \Leftrightarrow \neg \mathrm{Tr}(\ulcorner \mathrm{L} \urcorner)$$

On the other hand

$$\Gamma \vdash \mathrm{L} \Leftrightarrow \mathrm{Tr}(\ulcorner \mathrm{L} \urcorner)$$

This contradicts the consistency of Γ and completes the proof. □

16. Incompleteness theorems

In this section we reprove first incompleteness theorems and prove also second incompleteness theorem. We use method originally presented by Gödel in [Gödel, 1931].

Definition 16.1. Let γ be a predicate of formulas. Then a sentence G_γ such that

$$\mathrm{PRA} \vdash G_\gamma \Leftrightarrow \neg \mathrm{Pr}_\gamma(\ulcorner G_\gamma \urcorner)$$

is a *Gödel sentence* for γ.

Gödel sentences exist according to Theorem 12.6.

Theorem 16.2 [Metatheory][First incompleteness theorem]. *Let* Γ *be a collection of formulas of the language of* PRA *and let* γ *be a predicate of formulas. Assume that the following assertions hold.*

(1) Γ *is an extension of* PRA.

(2) $\Gamma \vdash \gamma(\ulcorner \Phi(\vec{n}) \urcorner)$ *for every formula* $\Phi(\vec{n})$ *in* Γ.

let G_γ *be a Gödel sentence for* γ. *If* Γ *is consistent, then* $\Gamma \nvdash G_\gamma$.

Proof. Since Γ contains all axioms of PRA, we infer that

$$\Gamma \vdash G_\gamma \Leftrightarrow \neg \mathrm{Pr}_\gamma(\ulcorner G_\gamma \urcorner)$$

Assume that Γ is consistent and $\Gamma \vdash G_\gamma$. On the one hand Corollary **??** implies that **HB1** holds for Γ and thus $\Gamma \vdash \mathrm{Pr}_\gamma(\ulcorner G_\gamma \urcorner)$. On the other hand $\Gamma \vdash G_\gamma \Leftrightarrow \neg \mathrm{Pr}_\gamma(\ulcorner G_\gamma \urcorner)$ and thus $\Gamma \vdash \neg \mathrm{Pr}_\gamma(\ulcorner G_\gamma \urcorner)$. Combining these two we obtain that Γ is inconsistent. This is contradiction. □

Definition 16.3. Let γ be a predicate of formulas. Then

$$\mathrm{Con}_\gamma \equiv \neg \mathrm{Pr}_\gamma(\bot)$$

is the *consistency sentence* for γ.

Theorem 16.4 [Metatheory][Formalized first incompleteness]. *Let* Γ *be a theory in the language of* PRA *and let* γ *be a predicate of formulas. Assume that the following assertions hold.*

(1) Γ *is an extension of* PRA.

(2) $\Gamma \vdash \gamma(\ulcorner \Lambda \urcorner)$ *for every sentence* Λ *in* Γ.

(3) $\mathrm{Pr}_\gamma(\phi)$ *is a* Σ_1*-formula.*

Then $\Gamma \vdash \mathrm{Con}_\gamma \Rightarrow G_\gamma$.

Proof. According to Corollary 14.2 Hilbert-Bernays conditions holds for Γ and γ. We use them to formalize Theorem 16.2.

On the one hand by **HB3** we infer

$$\Gamma \vdash \mathrm{Pr}_\gamma(\ulcorner G_\gamma \urcorner) \Rightarrow \mathrm{Pr}_\gamma\left(\ulcorner \mathrm{Pr}_\gamma(\ulcorner G_\gamma \urcorner)\urcorner\right)$$

and hence

$$\Gamma, \mathrm{Pr}_\gamma(\ulcorner G_\gamma \urcorner) \vdash \mathrm{Pr}_\gamma\left(\ulcorner \mathrm{Pr}_\gamma(\ulcorner G_\gamma \urcorner)\urcorner\right)$$

On the other hand, since Γ contains all axioms of PRA, we infer that

$$\Gamma \vdash \mathrm{Pr}_\gamma(\ulcorner G_\gamma \urcorner) \Rightarrow (\neg G_\gamma)$$

By virtue of **HB1** we have

$$\Gamma \vdash \mathrm{Pr}_\gamma\left(\ulcorner \mathrm{Pr}_\gamma(\ulcorner G_\gamma \urcorner)\urcorner \rightarrow (\neg \ulcorner G_\gamma \urcorner)\right)$$

Combining these two with **HB2** we derive

$$\Gamma, \mathrm{Pr}_\gamma(\ulcorner G_\gamma \urcorner) \vdash \mathrm{Pr}_\gamma(\neg \ulcorner G_\gamma \urcorner)$$

Since $\neg \ulcorner G_\gamma \urcorner = G_\gamma \rightarrow \bot$, we apply **HB2** to deduce

$$\Gamma, \mathrm{Pr}_\gamma(\ulcorner G_\gamma \urcorner) \vdash \mathrm{Pr}_\gamma(\bot)$$

Next we apply deduction theorem to infer

$$\Gamma \vdash \mathrm{Pr}_\gamma(\ulcorner G_\gamma \urcorner) \Rightarrow \mathrm{Pr}_\gamma(\bot)$$

The law of contraposition and the fact that $\Gamma \vdash \neg \mathrm{Pr}_\gamma(\ulcorner G_\gamma \urcorner) \Rightarrow G_\gamma$ imply that

$$\Gamma \vdash \mathrm{Con}_\gamma \Rightarrow G_\gamma$$

□

Corollary 16.5 [Metatheory][Second incompleteness]. *Let* Γ *be a theory in the language of* PRA *and let* γ *be a predicate of formulas. Assume that the following assertions hold.*

(1) Γ *is an extension of* PRA.

(2) $\Gamma \vdash \gamma(\ulcorner \Lambda \urcorner)$ *for every sentence* Λ *in* Γ.

(3) $\mathrm{Pr}_\gamma(\phi)$ *is a* Σ_1*-formula.*

If Γ *is consistent, then* $\Gamma \nvdash \mathrm{Con}_\gamma$.

Proof. According to Theorem 16.4 we have $\Gamma \vdash \mathrm{Con}_\gamma \Rightarrow G_\gamma$. Theorem 16.2 implies that if Γ is consistent, then $\Gamma \nvdash G_\gamma$. Hence if Γ is consistent, then $\Gamma \nvdash \mathrm{Con}_\gamma$. □

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